

Reading 9: Correlation & Regression

- Sample Cov (X, Y) = $\frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n-1}$
- Correlation Coefficient = $r_{XY} = \frac{\text{cov}_{XY}}{(s_X)(s_Y)}$ or $r = \frac{\text{cov}(X,Y)}{\sqrt{\text{var}(X)}\sqrt{\text{var}(Y)}}$
- t-test (for normally distributed variables) = $t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$ t distribution with (n - 2) deg. of freedom
- Linear Regression = $Y_i = b_0 + b_1X_i + \varepsilon_i$,
 - Intercept (b_0) = $\bar{b}_0 = \bar{y} - b_1\bar{x}$
 - Slope or regression coefficient = $b_1 = \frac{\text{cov}(x,y)}{\text{var}(x)}$ or $= \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2}$
- Standard Error of Estimate SEE = $S_E = \sqrt{\frac{\text{SSE}}{n-k-1}} = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y})^2}{n-k-1}}$
- Coefficient of Determination (R^2) = $= \frac{\text{SST} - \text{SSE}}{\text{SST}} = \frac{\text{RSS}}{\text{SST}}$ where, $0 \leq R^2 \leq 1$ (for single independent variable $R^2 = r^2$)
- $\text{SST} = \text{SSE} + \text{SSR}(\text{or RSS})$
- Hypothesis Testing:
 - Null and Alternative hypotheses
 - $H_0: b_1 = 0$ (no linear relationship)

- $H_1: b_1 \neq 0$ (linear relationship does exist)
- Test statistic = $t = \frac{\hat{b}_1 - b_1}{S_{b_1}}$
- Confidence Interval = $b_1 \pm t_c S_{b_1}$

9. ANOVA (Analysis of variance) =

ANOVA	SS	MSS	F
Regression df = k	$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$	$\frac{SSR}{k}$	$\frac{SSR/k}{SSE/(n-k-1)}$
Error df = n-k-1	$SSE = \sum_{i=1}^n (y_i - \hat{y})^2$	$\frac{SSE}{n-k-1}$	
Total df = n-1	$SST = \sum_{i=1}^n (y_i - \bar{y})^2$		

Source of Variability	DoF	Sum of Squares	Mean Sum of Squares
Regression (Explained)	1	RSS	MSR = RSS/1
Error (Unexplained)	n-2	SSE	MSE = SSE/n-2
Total	n-1	SST = RSS + SSE	

- F-Statistic or F-Test = $\frac{MSR}{MSE} = \frac{(\frac{RSS}{k})}{(\frac{SSE}{n-k-1})}$
(df numerator = k = 1)
(df denominator = n - k - 1 = n - 2)
- Prediction Intervals = $\hat{Y} \pm t_c S_f$
where $s_f^2 = s^2 \left[1 + \frac{1}{n} + \frac{(x-\bar{x})^2}{(n-1)s_x^2} \right]$ and $S_f = \sqrt{s_f^2}$

Reading 10: Multiple Regression & Issues in Regression Analysis

- $Y_i = b_0 + b_1X_{1i} + b_2X_{2i} + \dots + b_kX_{ki} + \varepsilon_i, i = 1, 2, \dots, n$
- Prediction equation = $\hat{Y}_i = \hat{b}_0 + \hat{b}_1X_{1i} + \hat{b}_2X_{2i} + \dots + \hat{b}_kX_{ki} + \varepsilon_i, i$
- Adjusted $R^2 = \bar{R}^2 = 1 - \left(\frac{n-1}{n-k-1} \right) (1 - R^2)$
- Breusch-Pagan test
 - H_0 = No conditional Heteroskedasticity exists
 - H_A = Conditional Heteroskedasticity exists
 - Test statistic = $n \times R^2_{\text{residuals}}$
- Durbin-Watson Test = $DW = \frac{\sum_{t=2}^T (\hat{\varepsilon}_t - \hat{\varepsilon}_{t-1})^2}{\sum_{t=1}^T \hat{\varepsilon}_t^2}$
 - For Large Sample size DW Statistic (d) = $d \approx 2(1 - r)$

Reject H_0 , conclude Positive Serial Correlation		Do not reject H_0		Reject H_0 , conclude Negative Serial Correlation	
0	d_L	Inconclusive	d_U	Inconclusive	$4-d_L$
			$4-d_U$		4

Reading 11: Time Series Analysis

- Linear Trend Models $y_t = b_0 + b_1 t + \varepsilon_t$
 - Predicted/fitted value of y_t in period $(T+1) = \hat{y}_{t+1} = \hat{b}_0 + \hat{b}_1(T+1)$
- Log-Linear Trend Models $y_t = e^{b_0 + b_1 t}$
- Autoregressive Time-Series Models:
 - First order autoregressive AR (1) $= x_t = b_0 + b_1 x_{t-1} + \varepsilon_t$
 - pth-order autoregressive AR (p) $= x_t = b_0 + b_1 x_{t-1} + b_2 x_{t-2} + \dots + b_p x_{t-p} + \varepsilon_t$
- Mean reverting level of $x_t = \frac{b_0}{1 - b_1}$
- Chain Rule of Forecasting:
 - One-period ahead forecast $= \hat{x}_{t+1} = \hat{b}_0 + \hat{b}_1 x_t$
 - Two-period ahead forecast $= \hat{x}_{t+2} = \hat{b}_0 + \hat{b}_1 x_{t+1}$
- Random Walks and Unit Roots:
 - Random Walk without drift $= x_t = x_{t-1} + \varepsilon_t$ where, $b_0 = 0$ and $b_1 = 1$.
 - Correcting Random Walk $= y_t = x_t - x_{t-1}$

- Random walk with a drift $= x_t = b_0 + x_{t-1} + \varepsilon_t$ where, $b_0 \neq 0$ and $b_1 = 1$
- By taking first difference $y_t = x_t - x_{t-1} = b_0 + \varepsilon_t$

- Using Dickey-Fuller Test $= x_t - x_{t-1} = b_0 + (b_1 - 1) x_{t-1} + \varepsilon_t$
- Smoothing Past Values with n-Period Moving Average $= \frac{x_t + x_{t-1} + x_{t-2} + \dots + x_{t-(n-1)}}{n}$
- Correcting Seasonality in Time Series Models:
 - For quarterly data $= x_t = b_0 + b_1 x_{t-1} + b_2 x_{t-4} + \varepsilon_t$
 - For monthly data $= x_t = b_0 + b_1 x_{t-1} + b_2 x_{t-12} + \varepsilon_t$
- ARCH model $= \hat{\varepsilon}_t^2 = \alpha_0 + \alpha_1 \hat{\varepsilon}_{t-1}^2 + \mu_t$ where μ_t is an error term
 - Predicting variance of errors in period $t+1 = \hat{\sigma}_{t+1}^2 = \hat{\alpha}_0 + \alpha_1 \hat{\varepsilon}_t^2$

Reading 12: Excerpt from 'Probabilistic Approaches, Scenario Analysis, Decision Tree & Simulations'

Reading 13: Currency Exchange Rates

- Bid-offer Spread = Offer price – Bid price

- Fwd rate = Spot Exchange rate + $\frac{\text{Forward points}}{10,000}$
- Forward premium/discount (in %) $= \frac{\text{spot exchange rate} - (\text{forward points}/10,000)}{\text{spot exchange rate}} - 1$
- To convert spot rate into forward quote:
 - Spot exchange rate $\times (1 + \% \text{ premium})$
 - Spot exchange rate $\times (1 - \% \text{ discount})$
- Covered interest rate parity:
 - $(1 + i_d) = S_{f/d} (1 + i_f) \left(\frac{1}{F_{f/d}} \right)$
 - $F_{f/d} = S_{f/d} \left(\frac{1 + i_f}{1 + i_d} \right)$
 - Using day count convention: $\left(1 + i_d \left[\frac{\text{Actual}}{360} \right] \right) = S_{f/d} \left(1 + i_f \left[\frac{\text{Actual}}{360} \right] \right) \left(\frac{1}{F_{f/d}} \right)$
 - $F_{f/d} = S_{f/d} \left(\frac{1 + i_f \left[\frac{\text{Actual}}{360} \right]}{1 + i_d \left[\frac{\text{Actual}}{360} \right]} \right)$
- Uncovered Interest Rate Parity :
 - $i_f - \% \Delta S_{f/d}^e = i_d$
 - $\% \Delta S_{f/d}^e = i_f - i_d$
 - Forward premium or discount:

- For one year horizon =

$$F_{f/d} - S_{f/d} =$$

$$S_{f/d} \left(\frac{i_f - i_d}{1 + i_d} \right) \cong S_{f/d} (i_f - i_d)$$

- Using day count convention:

$$F_{f/d} - S_{f/d} = S_{f/d} \left(\frac{\left[\frac{\text{Actual}}{360} \right]}{1 + i_d \left[\frac{\text{Actual}}{360} \right]} \right) (i_f - i_d)$$

7. Forward discount or premium as % of spot rate:

$$\frac{F_{f/d} - S_{f/d}}{S_{f/d}} \cong (i_f - i_d)$$

If uncovered interest rate parity holds

$$\bullet \quad \frac{F_{f/d} - S_{f/d}}{S_{f/d}} = \% \Delta S_{f/d}^e \cong (i_f - i_d)$$

8. Purchasing Power parity (PPP)

- $P_f = S_{f/d} \times P_d$
- $S_{f/d} = P_f / P_d$

9. Relative version of PPP = $\% \Delta S_{f/d} = \pi_f - \pi_d$

10. Ex ante version of PPP = $\% \Delta S_{f/d}^e = \pi_f^e - \pi_d^e$

11. Real Exchange Rate

$$q_{f/d} = \left(\frac{S_{f/d} P_d}{P_f} \right) = S_{f/d} \left(\frac{P_d}{P_f} \right)$$

$$q_{f/d} = S_{f/d} \left(\frac{CPI_d}{CPI_f} \right)$$

or

12. Fisher effect:

- $i_d = r_d + \pi_d^e$
- $i_f = r_f + \pi_f^e$
- $i_f - i_d = (r_f - r_d) + (\pi_f^e - \pi_d^e)$
- $(r_f - r_d) = (i_f - i_d) - (\pi_f^e - \pi_d^e)$

13. When both uncovered interest rate parity and ex-ante PPP hold:

- $(r_f - r_d) = \% \Delta S_{f/d}^e - \% \Delta S_{f/d}^e = 0$
- International Fisher Effect: $i_f - i_d = \pi_f^e - \pi_d^e$

14. When all the key international parity conditions are held at all times:
BOP = Current A/C + Capital A/C +
Official Reserve A/C = 0

15. Real exchange rate = $q_{f/d} = \bar{q}_{f/d} + [(r_d - r_f) - (\varphi_d - \varphi_f)]$

16. Interest Rate Differentials, Carry Trades and Exchange Rates

$$q_{L/H} = \bar{q}_{L/H} + (i_H - i_L) - (\pi_H^e - \pi_L^e) - (\phi_H - \phi_L)$$

17. Policy Rate under Taylor rule:

$$\bullet \quad i = r_n + \pi + \alpha(\pi - \pi^*) + \beta(y - y^*)$$

- Neutral real policy rate + Current inf rate + α (Inf gap) + β (Output gap)

18. Exchange rates using the Taylor Rule =
 $q_{USD/EUR} = \bar{q}_{USD/EUR} + (r_n^{EU} - r_n^{US}) + \alpha((\pi_{EU} - \pi^{*EU}) - (\pi_{US} - \pi^{*US})) + \beta((y_{EU} - y^{*EU}) - (y_{US} - y^{*US})) - (\phi_{EU} - \phi_{US})$

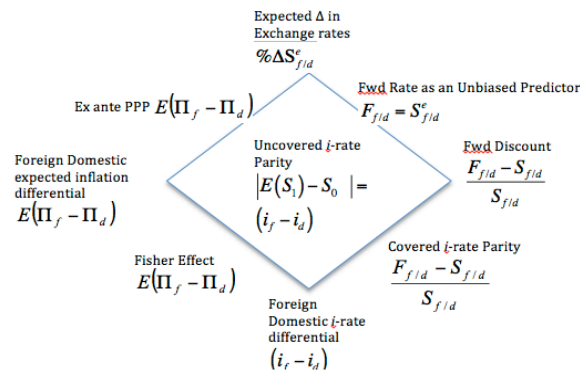
Reading 14: Economic Growth & The Investment Decision

1. Economic growth = Annual % Δ in real GDP or in real per capita GDP

$$2. \quad P = \text{GDP} \left(\frac{E}{\text{GDP}} \right) \left(\frac{P}{E} \right)$$

3. Expressing in terms of logarithmic rates:

- $(1/T) \% \Delta P = (1/T) \% \Delta \text{GDP} + (1/T) \% \Delta (E / \text{GDP}) + (1/T) \% \Delta (P / E)$
- $\% \Delta$ in stock MV = $\% \Delta$ in GDP + $\% \Delta$ in share of earnings (profit) in GDP



+ % Δ in the price-to-earnings multiple

4. A two-factor aggregate production function: $Y = A F(K, L)$
5. Cobb-Douglas Production Function = $F(K, L) = K^\alpha L^{1-\alpha}$
6. Under the Cobb-Douglas production function:
 - Marginal product of capital = $MPK = \alpha A K^{\alpha-1} L^{1-\alpha} = \alpha Y/K$
 - $\alpha Y/K = r \Rightarrow \alpha = r(K)/Y = \text{Capital income} / \text{Output or GDP}$
7. Output per worker or Average labor productivity (Y/L or y):
 - $\text{GDP/Labor input} = \text{TFP} \times \text{capital-to-labor ratio} \times \text{share of capital in GDP}$
 - Or $y = Y/L = A k^\alpha$
8. Contribution of Capital Deepening = Labor productivity growth rate – TFP
9. Contribution of Improvement in technology = Labor productivity growth rate – Capital Deepening
10. Growth Accounting based on Solow Approach = $\Delta Y/Y = \Delta A/A + \alpha \Delta K/K + (1-\alpha) \Delta L/L$
11. Labor productivity growth accounting equation

- Growth rate in potential GDP = LT g rate of labor force + LT g rate in labor productivity

12. Balanced or Steady State Rate of Growth in Neoclassical Growth Theory:

- Growth in physical capital stock = $\Delta K = sY - \delta K$

13. In the steady state:

- Growth rate of capital per worker = $\Delta k/k = \Delta y/y = \Delta A/A + \alpha \Delta k/k = \frac{TFP}{1-\alpha} \rightarrow \text{Steady state growth rate of labor productivity}$
- Growth rate of Total output = $\Delta Y/Y = \text{Growth rate of TFP scaled by labor force share} + \text{Growth rate in the labor force} = \frac{\theta}{1-\alpha} + n$
- Steady state Output-to-capital ratio = $\frac{Y}{K} = \left(\frac{1}{s}\right) \left[\left(\frac{\theta}{1-\alpha}\right) + \delta + n\right] = \Psi$
- Gross investment per worker = $\left[\left(\frac{\theta}{1-\alpha}\right) + \delta + n\right] k$
- Slope of straight line = $[\delta + n + \theta / (1-\alpha)]$

14. During the transition to the steady state growth path:

- Growth rates of output per capita = $\Delta y/y = \left[\left(\frac{\theta}{1-\alpha}\right) + \alpha s \left(\frac{Y}{K} - \Psi\right)\right] = \left(\frac{\theta}{1-\alpha}\right) + \alpha s (y/k - \Psi)$

- Capital-to-labor ratio = $\Delta k/k = \left[\left(\frac{\theta}{1-\alpha}\right) + s \left(\frac{Y}{K} - \Psi\right)\right] = \left(\frac{\theta}{1-\alpha}\right) + s (y/k - \Psi)$

15. Proportional impact of the saving rate change on the capital-to-labor ratio and per capita income over time:

$$\frac{k_{new}}{k_{old}} = \left[\frac{\left(\frac{Y}{K}\right)_{new}}{\left(\frac{Y}{K}\right)_{old}} \right]^{\frac{1}{\alpha-1}}$$

$$\frac{y_{new}}{y_{old}} = \left[\frac{k_{new}}{k_{old}} \right]^\alpha$$

16. Production function in the endogenous growth model = $y_e = f(k_e) = c k_e$

- Growth rate of output per capita = $\Delta y_e/y_e = \Delta k_e/k_e = s c - \delta - n$

Reading 15: Economics of Regulation

Reading 16: Inventories: Implications for Financial Statements & Ratios

1. End. Inventory = Beg. inventory + purchases – COGS
2. WAC per unit = $\frac{\text{Total cost of goods available for sale}}{\text{Total units available for sale}}$

3. LIFO Reserve = FIFO Inventory – LIFO Inventory
4. FIFO COGS = LIFO COGS – *Inc* in LIFO reserve = LIFO COGS – (End LIFO Reserve – Beg LIFO reserve)
5. N.I (FIFO) = Net Income (LIFO) + [Δ in LIFO reserve $\times$ (1 – tax rate)]
6. R.E (FIFO) = R.E (LIFO) + [LIFO reserve $\times$ (1 – tax rate)]
7. Tax liability (FIFO) = Tax liability (LIFO) + (LIFO reserve $\times$ tax rate)
8. Cumulative amount of Income Tax savings using the LIFO method e.g. in year 2015 = (LIFO Reserve at end of 2015 $\times$ tax rate in 2015)
9. Amount added to R.E e.g. in year 2015, if FIFO method is used instead of LIFO method = [LIFO Reserve at end of 2015 $\times$ (1 – tax rate in 2015)]
10. To convert B.S to FIFO:
 - Inventory is increased by LIFO Reserve.
 - Cash is decreased by (LIFO Reserve $\times$ Tax rate %).
 - Shareholders' equity (R.E) is increased by [LIFO Reserve $\times$ (1 – tax rate %)]

11. Inventory Turnover Ratio =
$$\frac{\text{Cost of goods sold}}{\text{Average Inventory}}$$
12. Days of Inventory on hand =
$$\frac{\text{No of days in period}}{\text{Inventory turnover ratio}}$$
13. Gross Profit margin =
$$\frac{\text{Gross Profit}}{\text{Total revenue}}$$
14. Net Profit margin =
$$\frac{\text{Net Income}}{\text{Total revenue}}$$
15. Return on Assets =
$$\frac{\text{Net Income}}{\text{Average total assets}}$$
16. Current Ratio =
$$\frac{\text{Current Assets}}{\text{Current Liabilities}}$$
17. Total Liabilities to Equity Ratio =
$$\frac{\text{Total Liabilities}}{\text{Total Shareholders' equity}}$$
18. Gross Profit due to LIFO Liquidation = Reported Gross profit – gross profit (without LIFO Liquidation) or
= No of units liquidated $\times$ (Replacement cost (RC) per unit – Historical cost per unit)
19. Inventory Valuation Rules under IFRS: Inventory is measured (carried on the B.S) at lower of cost and NRV.
20. NRV = Estimated Sales price – Estimated Selling costs and completion costs

21. Inventory Valuation Rules under U.S. GAAP: Inventory is measured (carried on the B.S) at lower of cost or market
 - Market is usually equal to RC, subject to upper and lower limits
 - Market cannot be $>$ NRV and Market cannot be $<$ (NRV – normal profit margin)
 - If $RC > NRV \rightarrow$ Market is NRV
 - If Replacement cost $<$ (NRV – normal profit margin) $\rightarrow$ Market is (NRV – normal profit margin)

Reading 17: Long Lived Assets: Implications for Financial Statements & Ratios

Adjusted Ratios for Capitalization:

1. Interest coverage ratio =
$$\frac{\text{EBIT}}{\text{Interest Expense}}$$
2. Adjusted Interest coverage ratio =
$$\frac{\text{EBIT} + \text{Dep exp (previously cap int cost)}}{\text{int exp} + \text{cap int}}$$
3. Adjusted Total Assets = Total Assets – cap int + related dep
4. Adjusted Net Profit Margin =
$$\frac{\text{NI} - \text{int exp} + \text{related dep}}{\text{total Rev}}$$
5. Adjusted EBIT = EBIT(Reported) + Amort. of previously cap int.

6. Adjusted EBITDA = Adj EBIT + dep exp + amort. Exp. or Adj. EBITDA = EBITDA (as reported)

7. Adj. Operating CF = Reported operating CF – cap int

8. Adj. Investing CF = Reported investing CF + cap int

Depreciation:

9. Dep Exp = $\frac{\text{Original cost} - \text{salvage value}}{\text{Depreciable or useful life}}$

10. Accelerated Dep (e.g. Double Declining) =

$$\text{Dep exp in year } x = \frac{2}{\text{Asset life in years}} \times \text{BV at beg of year } x$$

11. Units-of-Production method =

Dep exp in year x =

$$\left[\frac{\text{No of units produced in yr } x}{\text{No of units asset is expected to produce over its useful life}} \right] \times (\text{Cost} - \text{estimated salvage value})$$

Impairment of Long-Lived Tangible Assets (held for use):

12. Under IFRS impairment losses are recognized when:

- CA of asset > RA of asset where,
- RA is higher of (Fair value – costs to sell) and VIU
- VIU = discounted expected future CFs

(CA = carrying amount, RA = recoverable amount)

13. Under U.S. GAAP (two step test):

- Recoverability test: Asset's CA is not recoverable when CA of asset > Undiscounted expected future CFs
- Impairment loss measurement: When asset's CA is not recoverable, then Impairment loss = asset's CA – asset's Fair value

Revaluation of Long-Lived Assets

14. Historical Cost Model = Historical cost – Acc Dep / Amort – Impa loss

15. Revaluation model: Fair value – Acc dep or Amort – Impa losses

16. Fixed Asset Turnover Ratio

$$= \frac{\text{Total Rev}}{\text{Avg Net Fixed Assets}}$$

17. Estimated total useful life = Estimated age + Estimated remaining life

18. Estimated total useful life = Avg depreciable life = $\frac{\text{Historical cost}}{\text{Annual Dep exp}}$

19. Estimated age = Avg. age (yrs) =

$$\frac{\text{Acc Dep}}{\text{Annual Dep exp}}$$

20. Estimated remaining useful life

$$= \frac{\text{Net PPE}}{\text{Annual Dep exp}} = \frac{\text{Total Historical cost} - \text{Acc dep}}{\text{Annual Dep exp}}$$

21. Interest Rev = Lease receivable beg of the period × interest rate

22. Direct Finance Lease results when: PV of lease pmts = CA of the leased asset

23. Sales-Type Lease results when: PV of lease pmts > CA of the leased asset

Reading 18: Interoperate Investments

1. Summary of Accounting Treatment of Investments

	Income Statement (I.S)	Balance Sheet (B.S)	Statement of SH's Equity
Held-to-maturity	$i \text{ income} = \text{Market rate at purchase} \times \text{Initial fair value (FV) of a debt security}$ $\text{Ori income} = i \text{ pmt} - \text{Amort}$ $i \text{ pmt} = (\text{Coupon rate} \times \text{Par value})$ $\text{Amort} = i \text{ pmt} - i \text{ income}$ <u>If debt security is sold:</u> Realized g/l reported on I.S = SP – CV or Amort cost	- Initially, at FV (IFRS) or initial price paid (US GAAP) - Subsequently, reported at amort cost at the subsequent reporting date on B.S.	N/A
Held for trading security	$i \text{ income} = \text{Market rate} \times \text{Initial FV}$ - Unrealized g/l = FV at the end of Yr t – Amort Cost at end of Yr t <u>If debt security is sold:</u> - Realized g/l reported on I.S= SP – Recorded FV	- Initially, at FV. - Subsequently, at FV at subsequent reporting date on B.S.	
Designated at fair value	$i \text{ income} = \text{Market rate at purchase} \times \text{Initial FV}$ - Unrealized g/l = FV at the end of Yr t – Amortized Cost at end of Yr t <u>If debt security is sold:</u> - Realized g/l reported on I.S= SP – Recorded FV	- Reported at FV at the end of Yr t - Subsequently, at FV at the subsequent reporting date on B.S	
Available-for-sale	$i \text{ income} = \text{Market rate at purchase} \times \text{Initial Fv}$ <u>If debt security is sold:</u> - Cumulative unrealized g/l is removed from OCI and entire g/l recognized in P&l statement. - Where, Realized g/l in I.S = (SP – Recorded FV) + Unrealized g/l	- Reported at FV at the end of Yr t - Subsequently, at FV at the subsequent reporting date on the B.S.	Unrealized g/l (net of tax) = FV at end of Yr t – Amort Cost at end of Yr t Unrealized g/l (net of tax) is reported as OCI

2. Goodwill = Cost of acquisition – investor's share of the FV of the net identifiable assets

PP	Xxx
Less: (% of Ownership Interest × BV of Investee's Net Assets)	(xxx)
= Excess Purchase Price	Xxx
Less: Attributable to Net Assets:	
-Plant & Equipment (% of Ownership Interest × difference b/wBV & FV)	(xxx)
-Land (% of Ownership Interest × difference b/wBV & FV)	(xxx)
= Residual Amount (Treated as Goodwill)	Xxx

3. Amort. of Excess PP:

Investment in associate:

PP	Xxx
Add: Investor's share of Investee's NI (% of Ownership Interest × Investee's NI)	Xxx
Less: Div. received (% of Ownership Interest × Div. paid)	(xxx)
Less: Amort. of excess PP attributable to plant & equipment (Amount attributable to PP&E* ÷ Remaining life of PP&E)	(xxx)
= Balance in investment in Investee	Xxx

Where, *Amount attributable to Plant & Equipment = % of Ownership Interest of investor × (FV of P&E – BV of P&E)

Beg net assets	Xxx
Add: NI	Xxx
Less: Div. paid	(xxx)
= Ending net assets	Xxx
Investor's proportionate share of Investee's recorded net assets (% of Ownership Interest × Ending net assets)	Xxx

Add: Unamortized excess PP (Excess PP – Amount attributable to PP&E)	xxx
= Investment in Investee	xxx

Transactions with Associates:

4. Upstream Transactions:

Investor's share of Associate's reported NI (% of Ownership Interest × Reported net income)	xxx
Less: Amort. of excess purchase price	(xxx)
Less: Unrealized profit (% of Ownership Interest × Profit from upstream sale in Associate's NI)	(xxx)
= Equity Income to be reported as a line item on Investor's I.S*	xxx

- Balance in the investment in Associate to be reported at the end of year:

PP	xxx
Add: Equity income (as calculated above)*	xxx
Less: Div. received (% of Ownership Interest × Div paid)	(xxx)
= Value of Investment in Associate's company at the end of year	xxx

- Composition of Investment account:

Investor's proportionate share of Associate's net equity = [% of Ownership Interest × (beg BV of net assets) + (Reported NI of associate – Profit from upstream sale in Associate's NI) – Div. paid by the associate)]	xxx
Add: Unamortized excess PP (Excess PP – Amort. of excess PP)	xxx

5. Downstream Transactions

Investor's share of Associate's reported NI (% of Ownership Interest × Reported NI)	xxx
Less: Amort of excess PP	(xxx)
Less: Unrealized profit (% of Ownership Interest × Profit from the downstream sale in Associate's NI)	(xxx)
= Equity Income to be reported as a line item on Investor's I.S	xxx

Unrealized profit = % of goods unsold × Profit on the sale to investee
Investor's share of the unrealized profit =
Unrealized profit × % of goods unsold

Investor's share of associate's reported NI (% of Ownership Interest × Reported NI)	xxx
Less: Amort of excess PP	(xxx)
Add: Realized profit (% of goods unsold × Unrealized profit)	xxx
= Equity Income to be reported as a line item on Investor's I.S	xxx

Business Combinations

6. Merger = Company X + Company Y
= Company X
7. Acquisition = Company X + Company Y =
(Company X + Company Y)
8. Consolidation = Company X + Company Y = Company Z

Goodwill

9. Full Goodwill = Total FV of the Subsidiary – FV of subsidiary's identifiable net assets

10. Partial Goodwill Method:

- Goodwill = FV of acquisition – Acquirer's share of FV of all identifiable tangible and intangible assets, liabilities and contingent liabilities acquired
- Or
- Goodwill = Purchase price – parent's (acquirer's) proportionate share of the FV of subsidiary's identifiable net assets.

11. Under Acquisition method, the allocation of PP:

FV of the stock issued	xxx
Add: BV of Investee's net assets	xxx
= Excess PP	xxx
FV of the stock issued	xxx
Less: FV allocated to identifiable net assets	(xxx)
= Goodwill	xxx

12. Allocation of excess PP: Excess PPP =
Sum of diff b/w FV and BV of identifiable assets + Goodwill
13. Combined Assets & Liabilities (A&L) reported on Consolidated B.S under acquisition method: Consolidated B.S

under acquisition method = BV for A&L of Investor + FV for A&L acquired from Acquiree

14. Combined Paid-in Capital (PIC) = (FV of the stock issued to effect the transaction – Par value of the stock issued) + Additional PIC of investor
15. Minority Interest = % of subsidiary not owned by the Parent × Subsidiary's Equity
16. Value of non-controlling interest under full goodwill method = Non-controlling interest's proportionate interest in subsidiary × FV of subsidiary on acquisition date
17. Value of non-controlling interest under partial goodwill method = Non-controlling interest's proportionate interest in subsidiary × FV of the subsidiary's identifiable net assets on acquisition date

Goodwill Impairment:

18. Goodwill Impairment Test under IFRS:
- Impaired when CA of the Cash-generating Unit > RA of the Cash-generating Unit
 - Impairment loss = CA of Cash-generating Unit - RA of Cash-generating Unit where,
RA = Higher of Net SP and its VIU
Net SP = FV – costs to sell
VIU = PV of expected future CF of cash-generating unit

19. Goodwill Impairment Test under U.S. GAAP (Two Step Approach)

- Step 1: Goodwill Impairment Test
- Impaired when CV of Reporting Unit (including Goodwill) > FV of Reporting Unit (including Goodwill).
- Step 2: Measurement of Impairment loss = CV of Reporting unit's Goodwill - Implied FV of Reporting unit's Goodwill
- Where Implied FV of Reporting unit's Goodwill = FV of Reporting Unit – FV of Reporting unit's net assets

Reading 19: Employee Compensation: Post Employment & Share-Based

1. Under DC Plans: Pension exp = Co.'s annual contribution to plans adjusted for Δ in yr-end accruals
2. Funded Status = PV of DB obligations – FV of plan assets
3. Period pension cost of a Co.'s DB pension plan = Δ in Net pension liability or asset adjusted for employer's contributions
4. Net i exp = Discount rate $\times$ Net Pension liability
where Discount Rate = rate used to calculate PV of future pension benefits

5. Net i income = Discount rate $\times$ Net Pension asset
6. Net return on plan assets = Actual return on plan assets – (Plan assets $\times i$ rate)
7. Actuarial g/l = Actual return – (Plan assets $\times$ Expected return)
8. Total Periodic Pension Costs = Sum of components of periodic pension costs
 - Total periodic pension cost in a given period = Δ in Net pension liability or asset adjusted for employer contributions
 - Total Net periodic pension cost (End Funded Status* – Beg Funded Status*) – Employer Contribution
where *Pension liability is treated as a negative
9. Adjusted Total P&L pension exp (income)
 - = Current service costs + i costs + (-) actuarial losses (actuarial gains) + past service costs (or plan amendments) – (+) Actual return (loss) on plan assets
Or
 - = Reported Total P&L pension exp (income) + Expected return on plan assets – Actual return on plan assets

10. Adjusted Pre-tax Income:

- = Reported Pre-tax income + (Actual return on plan assets – Expected return on plan assets)
- Or
- = Reported Pre-tax income + Total reported pension and other post-retirement benefits - Current service costs - i exp component of pension cost + Actual return on plan assets

11. Adjusted Net Operating Exp = Reported Net operating exp – Total reported pension and other post-retirement benefits + Current service costs

12. Adjusted i Exp. = Reported i exp. + i exp. component of pension cost

13. Adjusted i and investment Income = Reported i and investment income + Actual return on plan assets

14. Compensation exp. = FV of stock on the Grant Date

16. Compensation exp recognized =
Unrecognized non-vested compensation exp
Remaining vesting period

Reading 20: Multinational Operations

1. Cumulative Translation Adjustment = CTA = Assets – Liabilities – Common Stock – Retained Earnings

2. Balance Sheet Exposure:

		Foreign Currency (FC)	
	B.S Exposure	Strengthens	Weakens
When assets translated at current X rate > liabilities translated at current X rate	Net Asset B.S exposure	+ve translation adj	-ve translation adj
When liabilities translated at current X rate > assets translated at current X rate	Net Liability B.S exposure	-ve translation adj	+ve translation Adj

(X = exchange)

3. Re-measurement Gain = NI – NI before re-measurement gain

4. Re-measurement Loss = NI – NI before Re-measurement loss

5. Rules For Translation Of A Foreign Subsidiary's FC Financial Statements (F.Ss) Into Parent's Presentation Currency Under IFRS & U.S. GAAP

Foreign Subsidiary's Functional Currency		
	FC	Parent's Presentation Currency
Translation Method:	Current Rate method	Temporal Method
X rate at which F.Ss are translated from foreign subsidiary's bookkeeping currency to parent's presentation currency.		
ASSETS <i>Monetary assets:</i> Cash, a/c receivables	Current rate	Current rate
<i>Nonmonetary Assets:</i> i) Measured at current value i.e. marketable securities & inventories measured at	Current rate	Current rate

Foreign Subsidiary's Functional Currency		
	FC	Parent's Presentation Currency
market value under the lower of cost or market rule. ii) Measured at historical costs e.g. PP&E	Current rate	Historical rate
LIABILITIES <i>Monetary liabilities:</i> a/c payable, LT debt, accrued exp., and deferred income taxes. <i>Nonmonetary liabilities:</i> i) measured at current value ii) not measured at current value i.e. deferred revenue	Current rate Current rate Current rate	Current rate Current rate Historical rate
EQUITY <i>Other than R.E</i> i.e. Common Stock <i>Retained Earnings (R.E)</i>	Historical rates Beg R.E + translated NI – div. translated at historical rate	Historical rates Beg R.E + translated NI – div. translated at historical rate
Revenues	Average rate	Average rate
EXPENSES <i>Most Expenses</i> <i>Expenses related to assets translated at historical X rate</i> e.g. COGS, Dep., & Amort. etc.	Average rate Average rate	Average rate Historical rate
NI	Average rate	Mixed (a mix of average rate & historical rate)
Exposure	Net Assets or Net Liabilities	Net <i>monetary</i> assets or Net <i>monetary</i> liabilities
Treatment of translation adj. in parent's consolidated F.Ss	Accumulated as a separate component of equity	Included as g/l in NI

	TEMPORAL METHOD:		CURRENT RATE METHOD
	Net Monetary Liability Exposure	Net Monetary Asset Exposure	
FC strengthens relative to parent's presentation currency	- Rev ↑ - Assets ↑ - Liabilities ↑ - NI ↓ - SH' equity ↓ - Translation loss	- Rev ↑ - Assets ↑ - Liabilities ↑ - NI ↑ - SH' equity ↑ - Translation gain	- Rev ↑ - Assets ↑ - Liabilities ↑ - NI ↑ - SH' equity ↑ +ve Translation adj.
FC weakens relative to parent's presentation currency	- Rev ↓ - Assets ↓ - Liabilities ↓ - NI ↑ - SH' equity ↑ - Translation gain	- Rev ↓ - Assets ↓ - Liabilities ↓ - NI ↓ - SH' equity ↓ - Translation loss	- Rev ↓ - Assets ↓ - Liabilities ↓ - Net Income ↓ - SH' equity ↓ -ve Translation adj.

6. Impact of Changing Exchange Rates on Exposure

	Foreign Currency	
	Strengthens	Weakens
CURRENT RATE METHOD:		
Net Assets	Gain	Loss
Net Liabilities	Loss	Gain
TEMPORAL METHOD:		
Net Monetary Assets	Gain	Loss
Net Monetary Liabilities	Loss	Gain

Hyperinflationary Economy

$$7. \text{ Restatement Factor} = \frac{\text{Current yr's price index}}{\text{Historical price index}}$$

$$8. \text{ Restated Capital Stock} = \frac{\text{Capital stock original value} \times \text{Current yr's price index or date of contribution, whichever is later}}{\text{Historical price index}}$$

$$9. \text{ Restated Revenue} = \text{Revenue original value} \times \frac{\text{Current yr's price index}}{\text{Avg. price index}}$$

$$10. \text{ Loss from holding beg balance in cash} = \frac{-\text{Beg balance in cash} \times \text{Current yr's price index} - \text{Historical price index}}{\text{Historical price index}}$$

$$11. \text{ Loss from increase in cash during the yr} = \frac{-\text{Increase in cash} \times \text{Current yr's price index} - \text{Avg price index}}{\text{Avg price index}}$$

$$12. \text{ Gain from holding note payable} = \frac{\text{Notes payable} \times \text{Current yr's price index} - \text{Historical price index}}{\text{Historical price index}}$$

$$13. \text{ Avg. effective tax rate} = \frac{\text{Tax Exp}}{\text{Pretax Accounting Profits}}$$

$$14. \text{ Organic sales growth} = \text{Net sales growth} + \text{Foreign X impact} + \text{Acquisition/Divestiture impact.}$$

Reading 21: Evaluating Quality of Financial Reports

- DSR (days sales receivable index) = $(\text{Receivables}_t / \text{Sales}_t) / (\text{Receivables}_{t-1} / \text{Sales}_{t-1})$
- GMI (gross margin index) = $\text{Gross margin}_{t-1} / \text{Gross margin}_t$
- AQI (asset quality index) = $[1 - (\text{PP\&E}_t + \text{CA}_t) / \text{TA}_t] / [1 - (\text{PP\&E}_{t-1} + \text{CA}_{t-1}) / \text{TA}_{t-1}]$
- SGI (sales growth index) = $\text{Sales}_t / \text{Sales}_{t-1}$
- DEPI (depreciation index) = $\text{Dep rate}_{t-1} / \text{Dep rate}_t$
where, $\text{Dep rate} = \text{Dep} / (\text{Dep} + \text{PP\&E})$
- SGAI (sales, general, and admin exp index) = $(\text{SGA}_t / \text{Sales}_t) / (\text{SGA}_{t-1} / \text{Sales}_{t-1})$
- Accruals = $(\text{Income before extraordinary items} - \text{Cash from operations}) / \text{TA}$
- LEVI (leverage index) = $\text{Leverage}_t / \text{Leverage}_{t-1}$ where, $\text{Leverage} = \text{Debt} / \text{Assets}$
- Earnings $_{t+1} = \alpha + (\beta_1 \times \text{Earnings}_t) + \varepsilon$
- Account receivable turnover = $(365 / \text{DSO})$

$$11. \text{Z-score} = \left(1.2 \times \frac{\text{Net WC}}{\text{TA}}\right) + \left(1.4 \times \frac{\text{R.E}}{\text{TA}}\right) + \left(3.3 \times \frac{\text{EBIT}}{\text{TA}}\right) + \left(0.6 \times \frac{\text{M.V of Equity}}{\text{B.V of liabilities}}\right) + \left(1.0 \times \frac{\text{Sales}}{\text{T.A}}\right)$$

Reading 22: Integration of Financial Statement Analysis

- DuPont Analysis:
 - $\text{ROE} = \text{Tax Burden} \times \text{Interest Burden} \times \text{EBIT margin} \times \text{TATO} \times \text{Financial Leverage}$
 - $\text{ROE} = \text{NI/EBT} \times \text{EBT/EBIT} \times \text{EBIT/Sales} \times \text{Sales/Assets} \times \text{Assets/Equity}$
 - $\text{ROE} = \text{Net profit margin} \times \text{asset turnover} \times \text{leverage}$
 - Adjusted Asset base = $\text{Adjusted Total Assets} = \text{Total Assets of the company} - \text{Investments in Associates}$
 - Adjusted NI = $\text{NI of Co} - \text{NI from Associates}$
 - Adjusted Tax Burden = $\frac{\text{NI} - \text{Equity income}}{\text{EBT}}$
 - Adjusted TATO = $\frac{\text{Rev}}{\frac{\text{Beg TA} - \text{Beg Eq. Invstmnt.} + \text{End TA} - \text{End Eq. Invstmnt.}}{2}}$

Accruals and Earnings Quality

- B.S based aggregate accruals
 - $\text{Aggregate Accruals}_t = \text{NOA}_t - \text{NOA}_{t-1}$
where, $\text{NOA}_t = \text{Net operating Assets}_t = \text{Op Assets}_t - \text{Op Liab}_t = [\{\text{TA}_t - (\text{Cash}_t + \text{ST invstmnt.}_t)\} - \{\text{Total liab}$

$_t - (\text{Total LT debt}_t + \text{Debt in current liab.})\}$

- B. S based Accruals Ratio = $\frac{(\text{NOA}_t - \text{NOA}_{t-1})}{(\text{NOA}_t + \text{NOA}_{t-1})/2}$

- CF based aggregate accruals:
 - Aggregate Accruals = $\text{NI}_t - (\text{CFO}_t + \text{CFI}_t)$
 - CF based Accruals Ratio = $\frac{[\text{NI}_t - (\text{CFO}_t + \text{CFI}_t)]}{(\text{NOA}_t + \text{NOA}_{t-1})/2}$
 - Op. CF before interest and taxes = $\text{Op. CF} + \text{cash } i \text{ paid} + \text{cash taxes paid}$
 - Op income adjusted for accounting Δ = $\text{Profit before } i\& \text{ taxes} + \text{amort. of goodwill}$
- Cash Return on Assets = $\frac{\text{Op.CF}}{\text{Avg T.A}}$
- Cash Flow to Reinvestment = $\frac{\text{Op.CF}}{\text{capital expenditures}}$
- Cash Flow to Total Debt = $\frac{\text{Op.CF before interest \& taxes}}{\text{T.D}}$
- Capacity to pay debt (in years) = $\frac{\text{T.D}}{\text{Op.CF} - \text{Capital Expenditures}}$
- CF Interest Coverage = $\frac{\text{Op.CF before } i\& \text{ taxes}}{i \text{ Paid}}$

Decomposition and Analysis of the Co's Valuation:

9. Parent Co. pro-rata share of subsidiary/affiliates = (Subsidiary's share price in FC × Shares held by Parent Co. × X-rate)/Parent Co. total market capitalization
10. Implied Value of Parent Co. (excl. subsidiary/affiliates) = Parent Co.'s Mkt Cap - Value of subsidiary/affiliate holdings
11. P/E ratio of Parent Co = $\frac{\text{Parent Co.'s mkt Cap}}{\text{NI of Parent Co.}}$
12. Implied P/E ratio of Parent Co. = $\frac{\text{Implied Value of Parent Co. (excluding subsidiary/affiliates)}}{\text{NI of Parent Co.} - \text{Equity Income from subsidiary/affiliates}}$
13. Discount to Benchmark = $\frac{\text{Benchmark's } \frac{P}{E} - \text{Parent Co. P/E}}{\text{Benchmark's P/E}}$
- Off-Balance Sheet Leverage from Operating Leases
14. Adj. Fin Lev = $\frac{\text{T.A} + \text{PV of lease payments}}{\text{T.E}}$
15. Adj. D-to-E ratio = $\frac{\text{T.D} + \text{PV of lease payments}}{\text{T.E}}$
16. Adj. *i*-coverage Ratio = $\frac{\text{EBIT} - \text{Dep exp} + \text{Rent Exp}}{\text{i exp} + \text{Assumed i exp on leases}}$

Reading 23: Capital Budgeting

1. Depreciable Basis = Purchase price + any Shipping or handling or installation costs
- Expansion Project
2. Initial Outlay = FCInv + NWCInv
NWCInv = Δnon-cash current assets – Δnon-debt current liabilities = ΔNWC
3. Annual after-tax operating cash flow = CF = (S – C – D) (1 – T) + D or CF = (S – C) (1 – T) + TD
4. Terminal year after-tax non-operating cash flow = TNOCF = Sal_T + NWCInv – T (Sal_T – B_T)
- Replacement Project
5. Initial Outlay = FCInv + NWCInv – Sal₀ + T (Sal₀ – B₀)
6. Annual after-tax operating cash flow (incremental)
- CF = (ΔS – ΔC – ΔD) (1 – T) + ΔD or
 - CF = (ΔS – ΔC) (1 – T) + TΔD
7. Terminal year after-tax non-operating cash flow = TNOCF = ΔSal_T + NWCInv – T (ΔSal_T – ΔB_T)
8. (1 + Nominal rate) = (1 + Real rate) (1 + Inf rate)
9. Profitability index = PI = 1 + (NPV/Initial investment)

when PI > 1, invest and when PI < 1, do not invest.

10. CAPM = $r_i = R_F + \beta_i [E(R_M) - R_F]$

Economic and Accounting Income

11. Accounting income = Rev – Exp

12. Economic Income = AT CF from investment + Δ in MV = AT CF from investment + (End MV – Beg MV)
OR
= AT CF from invstmnt. – (Beg MV – End MV) = AT CF from invstmnt. – Eco. Dep

13. Economic Profit (EP) = NOPAT – \$WACC where,
NOPAT = net operating profit after tax i.e. EBIT (1 – Tax rate)
EBIT = earnings before interest and taxes
\$WACC = dollar cost of capital = WACC × capital
Capital (after Year 1) = investment = Initial Investment – depreciation

14. MVA or NPV = $\sum_{t=1}^{\infty} \frac{EP_t}{(1+WACC)^t}$

15. Total value of Co. = original investment + NPV

16. Residual income (RI) = NI – Equity Charge

- RI_t = NI_t – (r_e × B_{t-1}) where,

- MVA = $\sum_{t=1}^{\infty} \frac{RI_t}{(1+r_e)^t}$

- Total value of Co. = NPV (PV of RI)
+ Original Equity investment +
Original Debt investment

Claims Valuation

17. Total value of Co. = value of liabilities +
value of equity

Reading 24: Capital Structure

1. $WACC = r_{WACC} = \left[\frac{D}{V} \right] \times r_d \times (1 - t) + \left[\frac{E}{V} \right] \times r_e$
2. Total value of Co. = $V = D + E$
3. WACC without taxes = $r_{WACC} = \left[\frac{D}{V} \times r_d \right] + \left[\frac{E}{V} \times r_e \right]$
4. Cost of Equity = $r_e = r_0 + (r_0 - r_d) \frac{D}{E}$
5. $V = D + E = \frac{\text{Interest payments on debt}}{\text{Cost of debt}} + \frac{(\text{EBIT} - \text{interest payments on debt})}{\text{Cost of equity}}$
6. According to MM proposition I: $V_L = V_U$
and $E = V - D$
7. According to MM proposition II:
 - Cost of Equity = $r_e = r_0 + (r_0 - r_d) \frac{D}{E}$

- $V = D + E = \frac{\text{Interest payments on debt}}{\text{Cost of debt}} + \frac{(\text{EBIT} - \text{interest payments on debt})}{\text{Cost of equity}}$

8. Systematic Risk = $\beta_a = \left[\frac{D}{V} \right] \beta_d + \left[\frac{E}{V} \right] \beta_e$
 - $\beta_e = \beta_a + (\beta_a - \beta_d) (D/E)$
9. AT cost of debt = BT cost of debt $\times (1 - \text{Marginal tax rate})$
10. MM Proposition I with Taxes: Co.'s value is maximized at 100% Debt
 - $V_L = V_U + (t \times D)$
 - Value of Unlevered(all equity)Co. = $(VU) = \frac{\text{EBIT}(1 - t)}{r_{WACC}}$
 - $V = D + E = \frac{\text{Interest payments on debt}}{\text{Cost of debt}} + \frac{(\text{EBIT} - \text{interest payments on debt})(1 - t)}{\text{Cost of equity}}$
11. MM Proposition II with Taxes: WACC is minimized at 100% Debt
 - $r_e = r_0 + (r_0 - r_d)(1 - t) \frac{D}{E}$
 $(r_0 - r_d)(1 - t) = \text{Slope coefficient} *$
 $*(r_0 - r_d)(1 - t) < (r_0 - r_d)$
 - WACC with taxes: $r_{WACC} = \frac{D}{V} \times r_d \times (1 - t) + \frac{E}{V} \times r_e$
12. Static trade-off theory of capital structure
 $V_L = V_U + tD - \text{PV}(\text{Costs of financial distress})$

Reading 25: Dividends & Share Repurchases Analysis

Share is sold just before it goes ex-dividend:

1. Cash flow from Sale = Sale price – capital gains tax owed on the sale = $P_w - (P_w - P_b)(T_{CG})$
where, P_w = price with the right to receive dividend
 P_b = purchase price where b is for buy
 T_{CG} = marginal tax rate on capital gains

Share is sold (after share goes ex-div.)

2. CF from Sale = Sale price – cap gains tax (owed on sale) + AT amount of div. = $P_x - (P_x - P_b)(T_{CG}) + D(1 - T_D)$
3. When $P_x = P_w$ then $P_w - (P_w - P_b)(T_{CG}) = P_x - (P_x - P_b)(T_{CG}) + D(1 - T_D)$
 $P_w - P_x = D \left[\frac{1 - T_D}{1 - T_{CG}} \right]$ or $\Delta P = D \left[\frac{1 - T_D}{1 - T_{CG}} \right]$
where, $\Delta P = \Delta$ in price when the stock goes from with div to ex-div
4. Double Taxation Method: $\text{ETR} = \text{Corp. tax rate} + \{(1 - \text{Corp. tax rate})(\text{Indiv. tax rate})\}$
5. Dividend imputation tax system: $\text{ETR} = \text{SH's Marginal Tax Rate}$
6. Split tax system: $\text{ETR} = \text{Corp. tax rate on div} + \{(1 - \text{Corp. tax rate on div.})(\text{personal tax rate})\}$

Payout Policies:

7. Stable Div. Policy

- Expected $\uparrow$ in Div. = $\uparrow$ in Earnings $\times$ Target payout ratio $\times$ Adj. factor
- Adj. factor = $1/\text{no. of yrs. over which adj. in div. will take place}$
- Expected Div = Last div. + (Expected $\uparrow$ in earnings $\times$ Target payout ratio $\times$ Adj. factor)

8. Residual Div. Policy

- Div. = Earnings – (Capital budget $\times$ Equity % in capital structure) or
- Div. = Zero, whichever is greater.

$$9. \text{ Div. Payout Ratio} = \frac{\text{Div.}}{\text{NI}}$$

$$10. \text{ Div. Coverage Ratio} = \frac{\text{NI}}{\text{Div.}}$$

$$11. \text{ FCFE Coverage Ratio} = \frac{\text{FCFE}}{[\text{Div.} + \text{Share Repurchases}]}$$

$$\text{FCFE} = \text{CFO} - \text{FCInv} + \text{Net Borrowings}$$

Reading 26: Corporate Performance, Governance & Business Ethics

Reading 27: Corporate Governance

$$1. \text{ Share Overhang} = \frac{\text{No. of shares represented by the Options}}{\text{Total no. of shares outstanding}}$$

Reading 28: Mergers & Acquisitions

$$1. \text{ Statutory Merger} = \text{Co. X} + \text{Co. Y} = \text{Co. X}$$

$$2. \text{ Subsidiary Merger} = \text{Co. X} + \text{Co. Y} = (\text{Co. X} + \text{Co. Y})$$

$$3. \text{ Consolidation} = \text{Co. X} + \text{Co. Y} = \text{Co. Z}$$

$$4. \text{ New shares issued by Acquirer} = \frac{\text{Mkt cap of Target}}{\text{stock price of Acquirer}}$$

$$5. \text{ Post-merger no. of shares outstanding} = \text{Acquirer's pre-merger total shares outstanding} + \text{new shares issued by Acquirer}$$

$$6. \text{ Post-Merger EPS} = \frac{\text{Acquirer's pre merger Earnings} + \text{Target's pre merger Earnings}}{\text{Post merger number of shares outstanding}}$$

$$7. \text{ Post merger P/E (if market is efficient)} = \frac{\text{Pre merger stock price of Acquirer}}{\text{Post merger EPS}}$$

$$8. \text{ No. of acquirer shares received by each shareholder (in target Co.)} = \text{No. of target shares he/she owns} \times \text{X ratio}$$

$$9. \text{ HHI} = \sum_i^n \left(\frac{\text{Sales or output of Firm } i}{\text{Total Sales or output of market}} \times 100 \right)^2$$

$$10. \text{ Unlevered NI} = \text{NI} + \text{Net Interest after-tax}$$

- Net interest after-tax = $(i_{\text{exp}} - i_{\text{income}}) \times (1 - \text{Tax rate})$ Or
- Unlevered NI = EBIT $\times$ (1 - tax rate)
- NOPLAT = Unlevered NI + Δ in deferred taxes

$$11. \text{ FCF} = \text{NOPLAT} + \text{NCC} - \Delta \text{ in Net WC} - \text{Capex}$$

$$12. \text{ FCF} = \text{NI} + \text{net interest after-tax} + \Delta \text{ in deferred taxes} + \text{net noncash charges} - \Delta \text{ in NWC} - \text{Capex}$$

Terminal Value:

$$13. \text{ Using constant growth formula} \\ \text{Terminal Value}_T = \frac{\text{FCF}_T(1+g)}{(\text{WACC}_{\text{adjusted}} - g)}$$

$$14. \text{ Using Market Multiple} \\ \text{Terminal Value}_T = \text{FCF}_T \times \frac{P}{\text{FCF}}$$

$$15. \text{ EV} = \text{MV of debt} + \text{MV of equity} - \text{cash \& cash equivalents}$$

$$16. \text{ Takeover Premium} = \text{takeover (deal price) per share (of target Co.)} - \text{current stock price of target Co.} = \frac{\text{DP} - \text{SP}}{\text{SP}}$$

$$17. \text{ Estimated takeover price of Target} = \text{Estimated stock price of Target based on Comparables} + \text{Estimated takeover premium}$$

- When takeover premium is given in %, Estimated takeover price of Target = (Estimated stock price of Target based on Comparables) $\times$ (1 + Takeover premium in %)

18. Target Shareholders' gain = Premium = $P_T - V_T$
 where,
 P_T = price paid for target company
 V_T = pre-merger value of target company

19. Acquirer's gain = Synergies – Premium = $S - (P_T - V_T)$

20. Post-merger value of the combined company = $V_A^* = V_A + V_T + S - C$
 where,
 V_A = pre-merger value of the acquirer
 C = cash paid to target SH i.e. cash paid = cash price paid per share of target co. $\times$ no. of shares outstanding of target co.

21. In Stock offer = $P_T = (N \times P_{AT})$
 where,
 P_T = price paid for target co.
 N = No. of new shares target receives
 P_{AT} = price per share of combined firm after merger announcement

Reading 29: Equity Valuation: Applications & Processes

1. Mispricing = $V_E - P = (V - P) + (V_E - V)$
 • $V_E - P$: Mispricing
 • $V - P$: True Mispricing
 • $V_E - V$: Valuation Error
 where, V_E = estimated value
 P = market price
 V = intrinsic value

2. Residual Income Model = $NI - (\text{cost of equity} \times \text{Beg value Equity})$

Reading 30: Return Concepts

- Dividend yield or investment income = (D_H / P_0)
- Price appreciation $R = (P_H - P_0) / P_0$
- $HPR = r = \{(D_H + P_H) / P_0\} - 1$ OR $r = \{(P_1 - P_0 + CF_1) / P_0\}$
- Expected Alpha = $\text{Exp. } R - \text{Req. } R$
- Realized Alpha (Ex-post alpha) = $(\text{Actual HPR}) - (\text{Contemporaneous Req. } R)$
- Expected HPR:
 - When an asset's intrinsic value $\neq$ market price, the investor expects to earn = $RR + \text{return from the convergence of price to value}$
 - When an asset's intrinsic value = price, the investor expects to earn RR only.

- $E(R_T) \approx r_T + \{(V_0 - P_0) / P_0\}$
 where, r_T = periodic required RoR,
- $\{(V_0 - P_0) / P_0\}$ = estimate of return from convergence over period

7. IRR:

- Intrinsic value = $D_1 / (k - g)$
- If asset (fairly priced), market price = intrinsic value: $k = (D_1 / P_0) + g$

8. Req ROE = $R_f + ERP$

9. GGM Intrinsic value = $D_1 / (k - g)$

Macroeconomic Model Estimates (Supply side models):

10. $ERP = [\{(1 + \text{EINFL}) (1 + \text{EGREPS}) (1 + \text{EGPE}) - 1\} + \text{EINC}] - \text{Expected } R_f$
 where EINFL = expected inf. (forecasted as) $\{(1 + \text{YTM of 20-yr T-bonds}) / (1 + \text{YTM of 20-yr TIPS})\} - 1$.
- EGREPS = expected growth rate in real EPS.
 - Real GDP growth rate = labor productivity growth + labor supply growth rate

Labor supply growth rate = population growth rate + increase in labor force participation rate

- EGPE = expected growth rate in P/E ratio. (For efficient markets $1 + \text{EGPE} = 1 + 0 = 1$.)
 - EINC = expected income component (includes dividend yield & reinvestment R)
11. CAPM: Required Return on share $i = \text{Current expected } R_f + B_i (\text{ERP})$
- where $\text{ERP} = \text{Expected } R \text{ on mkt portfolio} - R_f$
 - $\text{Beta} = \text{Cov of returns with mkt } R / \text{mkt portfolio var.}$
12. Adjusted Beta = $(2/3) (\text{Unadjusted beta}) + (1/3) (1.0)$
13. Beta Estimation for Thinly Traded Stocks and Nonpublic Companies
- $B_u \approx [1 / \{1 + (D/E)\}] \times B_e$
 - $B_e' \approx [1 + (D'/E')] \times B_u$
14. Multifactor Models = $r = R_f + (RP)_1 + (RP)_2 + \dots + (RP)_k$
- $RP_i = (\text{Factor sensitivity})_i \times (\text{Factor } RP)_i$.
15. The Fama-French Model (FFM): $r_i = R_f + B_i^{\text{market}} \times \text{RMRF} + B_i^{\text{size}} \times \text{SMB} + B_i^{\text{value}} \times \text{HML}$
- $\text{RMRF} = R_M - R_f$
 - $\text{SMB}(\text{small minus big}) = \text{Avg. } R \text{ on 3 small-cap portfolios} - \text{avg. } R \text{ on 3 large-cap portfolios.}$
 - $\text{HML}(\text{high minus low}) = \text{Avg. } R \text{ on 2 high Book-to-market portfolios} - \text{avg. } R \text{ on 2 low book-to-market portfolios.}$
16. Pastor-Stambaugh Model (PSM): $r_i = R_f + B_i^{\text{market}} \times \text{RMRF} + B_i^{\text{size}} \times \text{SMB} + B_i^{\text{value}} \times \text{HML} + B_i^{\text{Liq}} \times \text{LIQ}$
17. 5-factor BIRR Model: $r_i = \text{T-bill rate} + (\text{sensitivity to confidence risk} \times \text{confidence RP}) - (\text{sensitivity to time horizon} \times \text{time horizon RP}) - (\text{sensitivity to inf. risk} \times \text{inf. RP}) + (\text{sensitivity to business cycle risk} \times \text{business cycle RP}) + (\text{sensitivity to mkt. timing risk} \times \text{mkt. timing RP})$
18. Build-Up Approaches for Private Business Valuation: $r_i = r_f + \text{ERP} + \text{Size prem}_i + \text{Specific Co. prem}_i$
19. Bond yield Plus RP (BYPRP) cost of equity = $\text{YTM on the co.'s LT debt} + \text{RP}$
- Country Spread Model
20. ERP estimate = $\text{ERP for a developed mkt} + \text{Country prem.}$
- $\text{Country Prem.} = \text{yield on emerging mkt bonds (denominated in currency of developed market)} - \text{yield on developed mkt. govt. bonds}$
21. Cost of Capital = $\text{WACC} = \{D/(D+E)\}r_d (1 - \text{Tax rate}) + \{E/(D+E)\}r_E$
- Reading 31: The five competitive forces that shape strategy
- Reading 32: Your strategy needs a strategy
- Reading 33: Industry & company analysis
1. % of sales (specific geographic region) = $\text{Sales of a particular region} / \text{Total sales of a co.}$
 2. Co.'s projected Rev. growth = $\text{Projected mkt. share} \times \text{Projected sales of a given product mkt.}$
 3. Forecasted variable costs = % of rev. Or = $\text{Unit volume} \times \text{Unit variable costs}$
 4. $\text{COGS} = \text{Raw materials} + \text{Direct labor} + \text{Overhead (in producing the goods)}$
 5. Finance costs = $(\text{Fixed } i \text{ rate on debt} \times \text{Gross debt at beg. of period}) - (i \text{ income rate} \times \text{cash position at beg of period})$
 6. Gross debt = $\text{LT financial debt} + \text{ST financial debt} + \text{Accrued interest}$
 7. Net debt = $\text{Gross debt} - \text{Cash and cash equivalents}$

8. Effective i rate = $i \text{ exp} / \text{Avg gross debt}$
9. i rate on avg cash position = $i \text{ income} / \text{Avg cash position}$
10. i rate on avg. net debt = $\text{Net } i \text{ exp} / \text{Avg. net debt}$
11. Deferred tax asset/liability = Profit and loss (reported) tax amount – Cash tax amount
12. Projected A/C receivable = Forecasted annual sales (assuming all credit sales) $\times$ (Assumed DSO/ 365)
13. Projected inventory = Assumed COGS / Assumed Inventory TO ratio
14. ROIC= NOPLAT / Invested Capital = EBI / (Operating assets – Operating liab.)
15. ROCE = Op. profit / Capital employed (i.e. debt and equity capital)
16. Rev. loss for co. due to cannibalization of demand = Projected no. of units of product cannibalized by the new substitute product $\times$ Estimated ASP
- Where,
- Average selling price (ASP) = $\frac{\text{Company's estimated avg. rev.}}{\text{Company's estimated shipments of product}}$
 - No of units of a product cannibalized by the new substitute product =

- Expected no. of product shipments $\times$ % representation of each category (e.g. consumer & non-consumer) $\times$ Cannibalization factor for the category
17. Post cannibalization shipments = Pre-cannibalization shipments – Expected cannibalization
18. Post cannibalization revenue = Pre-cannibalization revenue – Estimated impact on rev. from cannibalization
19. Overall organic rev. growth = $[(1 + \text{volume growth}) (1 + \% \text{ of price/mix contribution to rev. growth})] - 1$
20. Construction of Pro Forma I.S:
- Sales
Less: COGS
= Gross profit
Less: Admin. exp.
Less: Distrib. Exp.
Add: Other income from operation
= EBIT
Add (Less): Other operating income (exp.)
Less: Finance costs & other financial exp.
= Profit before tax
Less: Income Tax
Add: Income from associates
= Profit from continuing operations

Add (Less): Profit (loss) from discontinued operations
= Net profit for the year
Less: Non-controlling interests
= Owners of the co.

21. EBITDA = EBIT + Dep. & amort. exp.

22. Forecasted CF Statement

CF from operating activities:

NI (profit after taxes)

Adj. to determine CF:

Add:

dep.

↓ in a/c receivable

↓ in inventory

↑ in a/c payable

Total adjustments

Net CF from operating activities

CF from investing activities:

Add: ↓ in plant and equipment

Net CF from investing activities

CF from financing activities:

↑ in notes payable

↑ in LTD

Less: Dividends paid

Net CF from financing activities

Forecasted ↑ in cash

23. Forecasted B.S

PP&E

Add: Investment in associates

Add: Other financial assets

Add: Deferred tax assets

= Total non-current assets

Inventories
 Add: Trade and other receivables
 Add: Cash & cash equivalents
 Add: Other current assets
 = Total current assets
 Total assets = Total non-current +
 Total current assets
 Share capital
 Add: Share premium
 Less: Treasury shares
 Add: Consolidated reserves + Net profit
 to co. owners
 Plus: Translation reserve
 +/- Profit or loss recorded in equity
 = Equity attributable to
 shareholders
 Plus: Non-controlling interest
 = Equity
 LT financial debt
 Add: Provision for employee benefits
 Add: LT provisions for liabilities and
 charges
 Add: Deferred tax liabilities
 = Total non-current liabilities
 ST financial debt and accrued interest
 Add: Trade and other payables
 Add: Income tax payable
 Add: ST provisions for liabilities and
 charges
 Add: derivative financial instruments
 Add: Liabilities held for sale
 = Current liabilities

24. FCFF:

Normalized operating profit

Less: Taxes
 = Normalized operating
 profit after tax
 Add: Dep. & amort.
 Δ in WC
 Less: Capital expenditures
 = FCFF

Reading 34: Discounted dividend valuation

- Asset's value is PV of its expected future
CFs i.e. $V_0 = \sum_{t=1}^n \frac{CF_t}{(1+r)^t}$
- RI = NI – (cost of equity × Beg. BV of
common equity)
- In RI Model: Value of stock = BVPS at t =
0 + PV of expected future residual
earnings
 - where, BVPS = common SHs' equity /
no. of common shares outstanding
 - RI model (assumes Clean Surplus
Accounting holds) i.e. $BV_t = BV_{t-1} +$
 $NI_t - Div_t$
- DDM
 - With Single HP = Value of Stock =
PV of expected Div. + PV of expected
Selling Price at the end of year one =
 $V_0 = \frac{D_1}{(1+r)^1} + \frac{P_1}{(1+r)^1}$
 - Value of stock for 2 years HP = $V_0 =$
 $\frac{D_1}{(1+r)^1} + \frac{D_2}{(1+r)^2} + \frac{P_2}{(1+r)^2}$
 - For n-HPs = $V_0 = \sum_{t=1}^n \frac{D_t}{(1+r)^t} + \frac{P_n}{(1+r)^n}$

- When HP is extended into indefinite
future: $V_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+r)^t}$

- Gordon Growth Model (GGM) = $V_0 =$
 $\frac{D_0 \times (1+g)}{r-g}$ or $\frac{D_1}{r-g}$
- GGM for Preferred stock (fixed rate
perpetual preferred stock) = $V_0 = \frac{D}{r}$
- GGM ERP = 1-yr. forecasted div. yield on
market index + consensus LT earnings
growth rate – LT govt. bond yield
- Actual value of a company's share = $V_0 =$
 $\frac{E_1}{r} + PVGO$
where,
 - PVGO = Sum of PV of expected
profitable opportunities of reinvesting
the earnings.
 E_1/r = no-growth value per share
 - When $P_0 = V_0$ then. $PVGO = P_0 - \frac{E_1}{r}$
 - Can be restated as $\frac{V_0}{E_1}$ or $\frac{P_0}{E_1}$ or $\frac{P}{E} =$
 $\left[\frac{1}{r} \right] + \left[\frac{PVGO}{E_1} \right]$
where, $1/r$ = value of P/E for no-
growth company.
 $PVGO/E_1$ = component of P/E value
that represents growth opportunities.
- Leading $\frac{P}{E}$ ratio = $\frac{P_0}{E_1} = \frac{D_1/E_1}{(r-g)} = \frac{(1-b)}{(r-g)}$

$$10. \text{ Trailing P/E ratio} = \frac{P_0}{E_0} = \frac{D_0(1+g)/E_0}{\frac{(1-b)(1+g)}{(r-g)}} =$$

$$11. \text{ GGM can be used to derive required RoR} \\ = r = \left[\frac{D_0(1+g)}{P_0} \right] + g = \left[\frac{D_1}{P_0} \right] + g$$

$$12. \text{ Two-Stage Div Discount Model} = V_0 = \sum_{t=1}^n \frac{D_t}{(1+r)^t} + \frac{V_n}{(1+r)^n}$$

where,

- $V_n = \frac{D_0 \times (1+g_S)^n (1+g_L)}{(r-g_L)}$
- $V_0 = \sum_{t=1}^n \left[\frac{D_0(1+g_S)^t}{(1+r)^t} \right] + \left[\frac{D_0 \times (1+g_S)^n \times (1+g_L)}{(1+r)^n (r-g_L)} \right]$

13. H-Model

- $V_0 = \frac{D_0 \times (1+g_L)}{(r-g_L)} + \frac{D_0 \times H \times (g_S - g_L)}{(r-g_L)}$ or
- $V_0 = \frac{[D_0 \times (1+g_L)] + [D_0 \times H \times (g_S - g_L)]}{(r-g_L)}$

where,

g_L = normal LT div. growth rate after year 2H

g_S = initial ST div. growth rate

H = half-life in years of the high-growth period i.e. high growth period = 2H years

$$14. \text{ Estimating Sustainable Growth Rate} = g = b \times \text{ROE}$$

- $g = \frac{\text{NI-Dividends}}{\text{NI}} \times \frac{\text{NI}}{\text{Sales}} \times \frac{\text{Sales}}{\frac{\text{Total Assets}}{\text{Shareholders' Equity}}}$

- g = PRAT i.e..
- g = profit margin (P) $\times$ retention rate (R) $\times$ asset turnover (A) $\times$ financial leverage (T)

$$15. \text{ ROE} = \frac{\text{NI}}{\frac{\text{Shareholders' equity}}{\text{Total Assets}}} = \frac{\text{NI}}{\text{Total Assets}} \times \frac{\text{Total Assets}}{\text{Shareholders' equity}}$$

$$= \frac{\text{NI}}{\text{Sales}} \times \frac{\text{Sales}}{\frac{\text{Total Assets}}{\text{Shareholders' equity}}} \times \frac{\text{Total Assets}}{\text{Shareholders' equity}} =$$

$$= \text{Net profit margin} \times \text{Asset Turnover} \times \text{Leverage}$$

Reading 35: Free cash flow valuation

$$1. \text{ PV of FCFF} = \text{Firm Value} =$$

$$\sum_{t=1}^{\infty} \frac{\text{FCFF}_t}{(1+WACC)^t}$$

$$2. \text{ WACC} = \left[\frac{\text{MV of D}}{\text{MV of D} + \text{MV of E}} \times \{r_d \times (1 - \text{Tax rate})\} \right] + \left[\frac{\text{MV of E}}{\text{MV of D} + \text{MV of E}} \times (r_e) \right]$$

$$3. \text{ PV of FCFE} = \text{Equity value} = \sum_{t=1}^{\infty} \frac{\text{FCFE}_t}{(1+r)^t}$$

$$4. \text{ Constant-Growth FCFF valuation Model} \\ \text{Firm Value} = V_0 = \frac{\text{FCFF}_1}{\text{WACC} - g} = \frac{\text{FCFF}_0 \times (1+g)}{\text{WACC} - g}$$

$$5. \text{ Constant-Growth FCFE valuation Model} \\ \text{Equity Value} = V_0 = \frac{\text{FCFE}_1}{r - g} = \frac{\text{FCFE}_0 \times (1+g)}{r - g}$$

$$6. \text{ Computing FCFF from NI} = \text{FCFF} = \text{NI} + \text{NCC} + \text{Int} \times (1 - \text{Tax rate}) + \text{Preferred stock div.} - \text{FCInv} - \text{WCInv}$$

a) When no LT assets are sold during the yr:

$$\text{FCInv} = \text{End gross PPE} - \text{Beg. gross PPE}$$

b) When LT assets are sold during the yr:

$$\text{FCInv} = \text{Capital expenditures} - \text{proceeds from sale of LT assets or}$$

$$\text{FCInv} = (\text{End. gross PPE} - \text{Beg. gross PPE}) - \text{Proceeds from sale of LT assets}$$

$$\text{WCInv} = \Delta \text{ in Current assets excl. cash \& cash equivalents} - \Delta \text{ in Current liab. excl. ST debt}$$

$$7. \text{ CF from operating activities} = \text{CFO} = \text{NI} + \text{NCC} - \text{WCInv.}$$

8. Computing FCFE from FCFF

- $\text{FCFE} = \text{FCFF} - \text{Int} \times (1 - \text{Tax rate}) - \text{preferred stock dividends} + \text{Net Borrowing} + \text{issuance of preferred stocks} - \text{redemption of preferred stock}$
- $\text{FCFE} = \text{NI} + \text{NCC} - \text{FCInv} - \text{WCInv} + \text{Net Borrowing} + \text{issuance of preferred stocks} - \text{redemption of preferred stock}$
- $\text{FCFE} = \text{CFO} - \text{FCInv} + \text{Net Borrowing} + \text{issuance of preferred stocks} - \text{redemption of preferred stock}$

- Total value of Equity (common) =
Total Firm value – Market value of
Debt – Preferred stock
9. Finding FCFF and FCFE from EBIT or EBITDA
- $FCFF = EBIT (1 - \text{Tax rate}) + \text{Dep} - \text{FCInv} - \text{WCInv}$
 - $FCFF = EBITDA (1 - \text{Tax rate}) + \text{Dep} (\text{Tax rate}) - \text{FCInv} - \text{WCInv}$
 - $FCFE = FCFF - \text{Int} (1 - \text{Tax rate}) + \text{Net borrowing} + \text{issuance of preferred stocks} - \text{redemption of preferred stock}$
10. Forecasted FCFF = Forecasted $[EBIT \times (1 - \text{Tax rate}) - \text{FCInv} - \text{WCInv}]$
11. Incremental fixed capital expenditures as a proportion of sales increases =

$$\frac{\text{Capital exp} - \text{Dep exp}}{\text{Increase in sales}}$$
12. Incremental working capital expenditures as a proportion of sales increases =

$$\frac{\text{Increase in WC}}{\text{Increase in sales}}$$
13. $FCFE = NI - (\text{FCInv} - \text{Dep}) - \text{WCInv} + \text{Net borrowing}$
 where,
 $\text{Net borrowing} = \text{DR} \times (\text{FCInv} - \text{Dep}) + \text{DR} \times (\text{WCInv})$
 $\text{FCFE} = NI - (\text{FCInv} - \text{Dep}) - \text{WCInv} + (\text{DR}) \times (\text{FCInv} - \text{Dep}) + (\text{DR}) \times (\text{WCInv})$
 Or
 $\text{FCFE} = NI - (1 - \text{DR}) \times (\text{FCInv} - \text{Dep}) - (1 - \text{DR}) \times (\text{WCInv})$

14. Modified Build-Up method to estimate real discount rate:

$$\begin{aligned} & \text{Country return (Real) [in \%]} \\ & +/ - \text{Industry Adjustment [in \%]} \\ & +/ - \text{Size Adjustment [in \%]} \\ & \underline{+/ - \text{Leverage Adjustment [in \%]}} \end{aligned}$$

$$\underline{\text{Required rate of return (real) [in \%]}}$$

15. Single-Stage FCFF and FCFE Model for International Valuation:

$$\begin{aligned} \text{Value of firm} = V_0 &= \frac{FCFF_0 \times (1 + g_{\text{real}})}{WACC_{\text{real}} - g_{\text{real}}} \\ &= \frac{FCFF_1}{WACC_{\text{real}} - g_{\text{real}}} \end{aligned}$$

$$\begin{aligned} \text{Value of Stock} = V_0 &= \frac{FCFE_0 \times (1 + g_{\text{real}})}{r_{\text{real}} - g_{\text{real}}} \\ &= \frac{FCFE_1}{r_{\text{real}} - g_{\text{real}}} \end{aligned}$$

16. Two-stage FCFF valuation model equation is:

- $\text{Firm Value} = \sum_{t=1}^n \frac{FCFF_t}{(1+WACC)^t} + \frac{FCFF_{n+1}}{(WACC-g)} \times \frac{1}{(1+WACC)^n}$
- Two-stage FCFE valuation model equation = Equity Value =

$$\sum_{t=1}^n \frac{FCFE_t}{(1+r)^t} + \frac{FCFE_{n+1}}{(r-g)} \times \frac{1}{(1+r)^n}$$

17. Excess Cash = Total Cash Available –

$$\left[\frac{\text{Total Assets of Firm} \times \text{Median level of Industry cash}}{\text{Median level of Industry Total Asset}} \right]$$

Reading 36: Market based valuation: Price and enterprise value multiples

1. Trailing P/E or Current P/E =

$$\frac{\text{Current Mkt Price per share}}{\text{most recent 4 Quarters' EPS}}$$
2. Forward P/E or Leading P/E or Prospective P/E =

$$\frac{\text{Current Market Price per share}}{\text{Next Year's Expected Earnings}}$$
3. Basic EPS =

$$\frac{\text{Total Earnings}}{\text{Wgtd Avg no.of shares actually o/s during the period}}$$
4. Diluted EPS =

$$\frac{\text{Total Earnings}}{\text{No.of shares o/s when holders of exercised their options to obtain common stock}}$$
5. Justified Forward P/E = $P_0/E_1 = \frac{D_1}{r-g} = \frac{1-b}{r-g}$
6. Justified Trailing P/E = $P_0/E_0 = \frac{D_0(1+g)/E_0}{r-g} = \frac{(1-b) \times (1+g)}{r-g}$ where
 P = price; E = earnings; D = dividends; r = required rate of return; and g = dividend growth rate
7. PEG ratio =

$$\frac{\text{Stock's P/E}}{\text{Expected Earnings Growth Rate in \%}}$$

8. Yardeni Model $CEY = CBY - (b \times LTEG) + \text{Residual}$

where,

CEY = current earnings yield on the mkt. index i.e. E/P .

CBY = current Moody's Investors Service A-rated corporate bond yield.

LTEG = consensus 5-year earnings growth rate forecast for the mkt index.

b = coefficient (measures weight, the mkt gives to 5-year earnings projections).

- By taking inverse: $\frac{P}{E} = \frac{1}{CBY - b \times LTEG}$

9. Own Historical P/E: Justified price = Benchmark value of own historical P/Es $\times$ Most recent EPS

10. Terminal Value (T.V) based on Fundamentals:

- T.V in yr n = (justified trailing P/E) $\times$ (forecasted earnings in year n)
- T.V in year n = (justified leading P/E) $\times$ (forecasted earnings in year n+1)

11. Terminal Value based on Comparables:

- T.V in yr n = (Benchmark trailing P/E) $\times$ (forecasted earnings in year n)
- T.V in yr n = (Benchmark leading P/E) $\times$ (forecasted earnings in year n+1)

12. $P/B = \frac{\text{Price per Share}}{\text{Book Value per Share}}$ where

- BVPS for equity shareholders = $\frac{TA - TL - P.S.}{\text{no. of common stock shares o/s}}$

$$= \frac{SH's' \text{ equity} - \text{total equity value claims that are senior to common stock}^*}{\# \text{ of C.S shares o/s}}$$

* It includes preferred stock and div. in arrears on preferred stock.

- BVPS for whole company = $\frac{\text{total assets} - \text{total liabilities}}{\text{number of shares outstanding}}$

13. Justified P/B = $P_0/B_0 = \frac{ROE - g}{r - g}$

14. Justified P/B based on RI model = $P_0/B_0 = 1 + \frac{\text{PV of expected future residual earnings}}{B_0}$

15. $P/S = \frac{\text{Price per share}}{\text{Annual net sales per share}}$
where Net Sales = Total Sales – returns – customer discounts

16. P/S (in terms of Gordon Growth Model =

$$\text{Justified P/S} = \frac{P_0}{S_0} = \frac{\left(\frac{E_0}{S_0}\right)(1-b)(1+g)}{(r-g)}$$

where, E_0/S_0 = Business's profit margin

17. $g = \text{Retention rate } (b) \times ROE$

$$g = b \times PM_0 \times \frac{\text{Sales}}{\text{Total Assets}} \times \frac{\text{Total Assets}}{\text{Shareholders' Equity}}$$

where, PM_0 = Profit Margin at $t = 0$

18. Price To Cash Flow

$$\begin{aligned} &= \frac{\text{Price per share}}{\text{Earnings plus noncash charges}} \text{ Or} \\ &= \frac{\text{Price per share}}{\text{Cash flow from Operations}} \text{ Or} \\ &= \frac{\text{Price per share}}{FCFE} \text{ Or} \\ &= \frac{\text{Price per share}}{EBITDA} \end{aligned}$$

19. Dividend Yield = $\frac{D}{P} = \frac{\text{Div per share}}{\text{Price per share}}$

- Trailing Div Yield = $\frac{\text{Dividend Rate}}{\text{Current mkt Price per share}}$
- Leading Div Yield = $\frac{\text{Forecasted Div per share over the next yr}}{\text{Current mkt. Price per share}}$

20. Div Yield (by using GGM) = Justified Div

$$\text{Yield} = \frac{D_0}{P_0} = \frac{r - g}{1 + g}$$

21. EV = MV of Common equity + MV of preferred stock* + MV of debt – Cash & Short-term Investments

- MV of Common equity = No. of shares o/s $\times$ Price per share
- Cash & Investments = cash, cash equivalents, short term investments etc.

*If minority interest exists and it is not included elsewhere, then it should be added back.

22. ROIC = $\frac{\text{Operating profit after tax}}{\text{Total invested Capital}}$

23. Total Invested Capital = TIC = MV of Common equity + MV of preferred stock + MV of debt

24. Earnings surprise $UE_t = EPS_t - E(EPS_t)$
 where,
 UE_t = unexpected earnings for quarter t
 EPS_t = reported/actual EPS for quarter t
 $E(EPS_t)$ = expected EPS for the quarter

- Percent Earning Surprise = $\frac{\text{Earnings Surprise}}{\text{Expected EPS}}$
- Scaled Earnings Surprise = $\frac{\text{Earnings Surprise}}{\text{S.D of analysts' earnings forecast}}$

25. Standardized Unexpected Earnings = SUE
 $t = \frac{EPS_t - E(EPS_t)}{\sigma [EPS_t - E(EPS_t)]}$
 where,
 EPS_t = reported/actual EPS for time t
 $E(EPS_t)$ = expected EPS for the time t
 $\sigma [EPS_t - E(EPS_t)]$ = S.D of $[EPS_t - E(EPS_t)]$ over some historical time period.

26. Relative strength indicator
 $= \frac{\text{Stock's performance}}{\text{Performance of an Equity Index}}$

27. Harmonic Mean = $X_H = \frac{n}{\sum_{i=1}^n \frac{1}{X_i}}$

Reading 37: Residual income valuation

1. End. BV of equity = Beg. BV of equity + Earnings – Div.

$$B_t = B_{t-1} + E_t - D_t \text{ \& } D_t = E_t - (B_t - B_{t-1}) = E_t + B_{t-1} - B_t$$

2. Residual Income (RI)

- = NI – Equity Charge = NI – (Equity Capital × Cost of Equity Capital)
- = NOPAT – Total Capital Charge = NOPAT – Debt Charge – Equity Charge = NOPAT – (AT cost of debt × Debt Capital) – (cost of equity × Equity Capital)
- RI (with preferred stock) = NI – Equity Charge – Preferred Stock Div
- RI = (ROIC – Effective Capital Charge) × Beg. Capital

3. EVA = NOPAT – (C% × TC)

Or = $[EBIT (1 - t)] - (WACC \times \text{invested capital})$
 where,
 C% = cost of capital
 TC = Total capital
 $WACC \times \text{invested capital}$ = dollar cost of capital
 Invested capital = net WC + net fixed assets = BV of LT debt + BV of equity

4. MVA = MV of Co. – Accounting BV of total capital = MV of Co – (BV of Debt + BV of Equity)

5. RI model

- $RI_t = E_t - (r \times B_{t-1}) = (ROE - r) \times B_{t-1}$
- Two components of intrinsic value of stock/equity

i. Current BV of Equity that is B_0

ii. PV of expected future RI $\left\{ \frac{RI_1}{(1+r)^1} + \frac{RI_2}{(1+r)^2} + \frac{RI_3}{(1+r)^3} + \dots \right\}$

- $V_0 = B_0 + \sum_{t=1}^{\infty} \frac{RI_t}{(1+r)^t} = B_0 + \sum_{t=1}^{\infty} \frac{E_t - rB_{t-1}}{(1+r)^t}$
- $V_0 = B_0 + \left\{ \frac{RI_1}{(1+r)^1} + \frac{RI_2}{(1+r)^2} + \frac{RI_3}{(1+r)^3} + \dots \right\}$

6. RI Model (general) = $V_0 = B_0 + \sum_{t=1}^{\infty} \frac{(ROE_t - r) \times B_{t-1}}{(1+r)^t}$

7. Justified $\frac{P}{B} = \frac{P_0}{B_0} = \frac{ROE - g}{r - g} = 1 + \left(\frac{ROE - r}{r - g} \right)$
 where, Justified Price is the stock's Intrinsic value i.e. $P_0 = V_0$
 BV of equity = B_0 = Total assets – total liab.

8. Tobin's q = $\frac{\text{MV of Debt and Equity}}{\text{Replacement cost of Total Assets}}$

9. Single-Stage RI Valuation = $V_0 = B_0 + \left(\frac{ROE - r}{r - g} \right) \times B_0$

- Implied Growth rate in RI = $g = r - \left[\frac{B_0(ROE - r)}{V_0 - B_0} \right]$

10. Multi-Stage RI Valuation = $V_0 = B_0 + (\text{PV of interim high-growth RI}) + (\text{PV of continuing RI})$

- PV of continuing RI in year T-1

$$= \frac{RI_T}{(1+r-\omega)(1+r)^{T-1}}$$
 where, ω = persistence factor, $0 \leq \omega \leq 1$
- Assumptions about Continuing RI:
 - RI is at +ve level currently and will persist at this level in the future indefinitely:
 PV of continuing RI in year T-1

$$= \frac{RI_T}{1+r-\omega} = \frac{RI_T}{1+r-1} = \frac{RI_T}{r}$$
 - RI will become 0 from the terminal yr forward.
 PV of continuing RI in year T-1

$$= \frac{RI_T}{(1+r-\omega)(1+r)^{T-1}} = \frac{RI_T}{(1+r-0)(1+r)^{T-1}} = \frac{RI_T}{(1+r)^{T-1}}$$
 - RI declines to 0 as ROE approaches r over time (& RI will become 0 eventually). i.e. $0 \leq \omega \leq 1$
 PV of continuing RI in year T-1 =

$$\frac{RI_T}{(1+r-\omega)(1+r)^{T-1}}$$
 - RI declines to long-run mean level of mature industry.

Where premium over book value is assumed at the end of time horizon T ($P_T - B_T$), current value (V_0) =

$$B_0 + \sum_{t=1}^T \frac{(ROE_t - r)B_{t-1}}{(1+r)^t} + \frac{P_T - B_T}{(1+r)^T}$$

Where $P_T = B_T \times$ (forecasted P/B ratio)

Reading 38: Private company valuation

1. Capitalized CF to the firm = $V_f = \frac{FCFF_1}{WACC - g_f}$
 - Value of Equity = $V_f - \text{MV of Debt}$
 - To value Equity directly = $V = \frac{FCFF_1}{r - g}$
2. Excess Earnings or RI = Normalized earnings – [(RR on WC × value of WC) + (RR on fixed assets × value of fixed assets)]
3. MVIC = MV of Debt + MV of Equity
4. Calculation of Lack of Control Discount =

$$DLOC = 1 - \frac{1}{1 + \text{Control Premium}}$$
5. Lack of Marketability Discount =

$$DLOM = \frac{\text{Value of "at-the-money" put option}}{\text{value of stock before any DLOM}}$$
6. Total Discount = $[1 - (1 - DLOC \text{ in } \%) \times (1 - DLOM \text{ in } \%)]$

Reading 39: Private real estate investments

1. Rent of Net Lease = Gross rent – Op. exp.
2. Lease Rent = Min. rent + % of sales rev above a certain level
 - In case of natural break-point, Lease Rent = % of sales rev. above a certain level × Tenant's sales

3. Implied land value = Value after construction – Cost to construct a building
4. Appraised value of a property

$$= \frac{NOI}{\text{Capitalization rate}}$$
 where,
 - NOI = Net operating income for the subject property
 - Gross potential income = Rental income at full occupancy + Other income
 - Effective gross income = Gross potential income – Vacancy and collection loss
 - NOI = Effective gross income – Op. exp. OR
 - NOI = Gross potential income – Estimated vacancy losses – Estimated collection losses – Insurance – Property Taxes – Utilities - Repairs and maintenance exp.
5. Capitalization rate = $\frac{\text{First-year NOI}}{\text{Property value}}$
6. Discount rate = Cap rate + Growth rate
7. Value of property = NOI / (discount rate – Growth rate)
8. Capitalization rate =

$$\frac{NOI}{\text{Sale price of comparable property}}$$
9. Reciprocal of the cap rate = $\frac{\text{Price}}{\text{Current NOI}}$

$$10. \text{ All Risks Yield} = \text{ARY} = \frac{\text{Rent}}{\text{Recent sales prices of comparables}}$$

$$11. \text{ Mkt. value} = \frac{\text{Rent}}{\text{ARY}}$$

12. Stabilized NOI i.e. (NOI of a non-renovated property or post-renovated NOI of a property)

- Post-renovation Value = Post-renovation NOI of a property / (Discount rate – growth rate)
- Loss in income due to renovation = Post-renovation NOI of a property - NOI of a property during renovation time-period
- PV of the lost income → Loss in value = Loss in income due to renovation / (1 + discount rate)
- Value of a property = Post-renovation Value - Loss in value OR
- Value of a property = {NOI of a property during renovation time-period + [Post-renovation NOI of a property (1 + growth rate)] / (Discount rate – growth rate)} / (1 + discount rate)

13. Avg. no. of months vacant until the lease is renewed = Lease Non-renewal probability × No. of months vacant if not renewed

14. Vacancy rate = Avg. no. of months vacant until the lease is renewed / (Lease term +

Avg. no. of months vacant until the lease is renewed)

$$15. \text{ Gross income multiplier} = \text{GIM} = \frac{\text{Sales Price}}{\text{Expected G.I. of property in 1st yr after sale}}$$

16. Total capital value = PV of income until the rent review + PV of what the property could be sold for at rent review (PV of estimated rental value or ERV)

17. Under Layer Method Value of a property = PV of current contract rent in perpetuity + PV of expected incremental rent after the rent review

18. When effective age of property < its economic life, Physical deterioration = % worn out = Effective age / Economic life

19. Incurable depreciation deduction = (Replacement cost + Developer's profit – curable depreciation costs) × Physical deterioration

20. Amount of functional obsolescence = Income loss due to the functional obsolescence / cap rate

21. Amount of locational obsolescence associated with building only = Total Loss in the value - Loss in land value

22. Cost Approach Calculations

- Total depreciation = Physical deterioration + functional obsolescence + Locational obsolescence + Economic obsolescence
- Depreciated building value = (Replacement cost + Developer's profit) – Total depreciation
- Final Appraisal value → Estimated value of the property = Depreciated building value + Land value

23. Valuation in an international context when land and building are valued separately:

- Income to the building = NOI – assumed land lease payment
- PV for building = Income to the building / Cap rate or discount rate
- Total value = PV for Building + Value of the land (from sales comparison approach)

24. The NCREIF Property Index (NPI):

- Total Return of individual Property = $\frac{\text{NOI} - \text{Capex} + (\text{Ending MV} - \text{Beg. MV})}{\text{Beg. MV}}$
- Income return = Cap rate = NOI / beginning value
- Amount of cash flows available each quarter = NOI – Capex
- Capital return = $\frac{(\text{Ending market value} - \text{Beginning market value}) - \text{Capex}}{\text{Beginning market value}}$
- Total Index Return = Value-Weighted average return for individual properties

25. Private market real estate debt

- $\text{Loan-to-value} = \text{Loan} / \text{value of the property}$
- $\text{Max. loan amount based on LTV ratio} = \text{LTV ratio (in \%)} \times \text{Appraisal value of property (in \$)}$
- $\text{Debt service coverage ratio} = \text{NOI} / \text{Debt service}$
- $\text{Debt service} = \text{Interest} + \text{Principal payments on the mortgage.}$
- $\text{Principal payments} = \text{Part of the loan payment that amortizes the loan over the loan term.}$
- $\text{Max. debt service based on DSCR} = \text{NOI} / \text{DSCR}$
- When the loan is interest-only, $\text{Max. loan amount based on DSCR} = \text{Max. debt service based on DSCR} / \text{Debt interest rate}$

26. Equity dividend rate or Equity yield rate = $\text{Cash flow} / \text{Equity}$

where, $\text{Cash flow} = \text{NOI} - \text{Debt Service}$

$\text{Equity} = \text{Price} - \text{Mortgage}$

27. Calculating Leveraged IRR:

CF received by the equity investor from the sale = $\text{Sale price} - \text{Mortgage balance}$
 $\text{PV} = - \text{Initial investment}$
 $\text{PMT} = \text{Cash flow}$
 $n = \text{Holding period}$
 $\text{FV} = \text{Cash flow received from sale}$
 $\text{CPT} \rightarrow \text{I/Y} \rightarrow \text{Leveraged IRR.}$

28. Calculating Unleveraged IRR:

Cash flow received by the equity investor from the sale = $\text{Sale price} + \text{NOI in the 1}^{\text{st}} \text{ year}$
 $\text{PV} = - \text{Initial investment}$
 $\text{PMT} = \text{NOI in the 1}^{\text{st}} \text{ year}$
 $n = \text{Holding period}$
 $\text{FV} = \text{Sale price}$
 $\text{CPT} \rightarrow \text{I/Y} \rightarrow \text{Unleveraged IRR.}$

Reading 40: Publicly traded real estate securities

1. $\text{Rent paid by Tenants} = \text{Net rent} + \text{Proportionate share of the common area costs of the mall (based on space leased)}$
2. $\text{NAVPS} = (\text{MV of R.E Co.'s assets} - \text{MV of R.E Co.'s liab.}) / \text{\# of shares outstanding}$
3. $\text{Appraised value} = \text{NOI} / \text{Cap rate}$
4. Estimating NAVPS:
 - $\text{Pro forma cash NOI} = \text{NOI} - \text{Non cash rents}^* + \text{Adj. for full impact of acquisitions}$
 - $\text{*Non-cash rent} = \text{Avg. contractual rent over the leases' terms} - \text{Cash rent actually paid.}$
 - $\text{Estimated future expected cash NOI} = \text{Pro forma cash NOI} + \text{Expected growth in NOI}$

- $\text{Estimated value of operating real estate} = \text{Estimated future expected cash NOI} / \text{Cap rate}$
- $\text{Estimated gross asset value} = \text{Estimated value of operating real estate} + \text{BV of Cash \& equivalents} + \text{BV of Land held for future development} + \text{BV of a/c receivables} + \text{BV of Prepaid/other assets}$
- $\text{Net asset value} = \text{Estimated gross asset value} - \text{Total debt} - \text{Other liab. (but not deferred taxes)}$
- $\text{NAVPS} = \text{Net asset value} / \text{\# of shares outstanding}$

5. $\text{P/FFO} = \text{Current stock prices} / \text{Yr-ahead estimated FFO}$

- $\text{FFO} = \text{Net earnings} + \text{Dep. Exp. on R.E} + \text{Deferred tax charges} - \text{g/l from sales of property and debt restructuring} + \text{Losses on sales of property and debt restructuring OR}$
- $\text{FFO} = \text{EBITDA} - \text{Interest Expense}$

6. $\text{P/AFFO} = \text{Current stock prices} / \text{Yr-ahead estimated AFFO}$

- $\text{AFFO} = \text{FFO} - \text{Non cash rent}^* - \text{Recurring Maintenance type Capital expenditures} - \text{Leasing costs (i.e. leasing agent's commissions} - \text{Tenants' improvement allowances)}$
- $\text{*Non-cash rent} = \text{Straight-line rent} - \text{Cash rent paid during the period}$

7. Estimated Value of a REIT Co. in yr N

$$= (P/FFO \text{ of overall REIT group for yr N}) \times \text{REIT co.'s expected FFO in yr N}$$
 or

$$= (P/AFFO \text{ of overall REIT group for yr N}) \times \text{REIT co.'s expected AFFO in yr N}$$

Reading 41: Private equity valuation

1. PIC:
 - PIC = Cumulative capital (CC) called down
 - PIC Multiple = PIC / CC
2. DPI = Sum of distb. / CC called down (or PIC)
3. RVPI = NAV after distb. / CC called down (or PIC)
4. TVPI = DPI + RVPI
5. Mgmt. fees = % fee $\times$ PIC
6. Carried interest = % $\times$ (NAV before distb. – CC) in year when NAV before distb. Is first $>$ CC
 Thereafter, Carried interest = % $\times$ Δ NAV before distb.
7. NAV before distb._t = NAV after distb._{t-1} + called down capital_t – mgmt fees_t + operating results_t

8. NAV after distb._t = NAV before distb._t – carried interest_t – distb._t
9. CF for Gross IRR_t = – Capital called down at the beg of period_{t+1} + Op. result_{t-1}
10. CF for Net IRR_t = – Capital called down at the beg of period_{t+1} + Op. result_{t-1} – mgmt. fees_t – carried interest_t

Venture Capital Method:

11. General Case: NPV Method

- Step 1: $POST = V / (1 + r)^t$
- Step 2: $PRE = POST - I$
- Step 3: $F = I / POST$
- Step 4: $y = x [F / (1 - F)]$
- Step 5: $p_1 = I / y$

12. Alternative Method using IRR:

- Step 1: $W = I (1 + r)^t$
- Step 2: $F = W / V$
- Step 3: $y = x [F / (1 - F)]$
- Step 4: $p_1 = I / y$
- Step 5: $POST = I / F$ or $p_1 \times (x + y)$
- Step 6: $PRE = POST - I$ or $PRE = p_1 \times x$

13. NPV Method with multiple (two) rounds of financing:

Step 1: Compound interest between dates:

- T_1 and $T_2 = (1 + R_1)$
- T_2 and $T_3 = (1 + R_2)$

- Step 2: $POST_2 = V / (1 + R_2)$
- Step 3: $PRE_2 = POST_2 - I_2$
- Step 4: $POST_1 = PRE_2 / (1 + R_1)$

- Step 5: $PRE_1 = POST_1 - I_1$
- Step 6: $F_2 = I_2 / POST_2$
- Step 7: $F_1 = I_1 / POST_1$
- Step 8: $y_1 = x_1 [F_1 / (1 - F_1)]$
- Step 9: $p_1 = I_1 / y_1$
- Step 10: $x_2 = x_1 + y_1$
- Step 11: $y_2 = x_2 [F_2 / (1 - F_2)]$
- Step 12: $p_2 = I_2 / y_2$

14. Accounting for Risk in Venture Capital

- By adjusting Discount Rate = $r = \left(\frac{1+r}{1-q} \right) - 1$

15. By adjusting terminal value using Scenario analysis = Adj. TV = (% prob. scenario 1 $\times$ expected E $\times$ expected P/E multiple) + (% prob. scenario 2 $\times$ expected E $\times$ expected P/E multiple) + ... + (% prob. scenario n $\times$ expected E $\times$ expected P/E multiple)

Reading 42: A primer on commodity investing

1. Value of the min. price Δ = U.S. \$ and cent-denom. tick $\times$ Contract size (or point value)
2. Future Price of a Commodity = $F_{0,T} = S_0 e^{rT}$
3. Future Price of a Commodity (with storage cost & convenient yield) = $F_0 = S_0 e^{(r+U-Y)T}$
4. Spot price return = $R_S = (S_t - S_{t-1}) / S_{t-1}$

5. Roll return = $R_T = (F_{t,T} - F_{t-1,T}) / F_{t-1,t} = F_{t,T} - S_t / S_t$
6. Commodity's Total R = Spot R + Roll R + Collateral R + Rebalancing R
7. Excess return Index = Futures R = Spot R + Roll R
8. Total return Index = Total R = Collateral R + Futures R = Collateral R + Spot R + Roll R

Reading 43: The term structure & interest rate dynamics

1. Discount Factor = $P(T) = \frac{1}{[1 + \text{spot rate}]^T} = \frac{1}{[1 + r(T)]^T}$
2. Forward pricing model: $P(T^* + T) = P(T^*) \times F(T^*, T)$
3. Forward rate model = $[1 + r(T^* + T)]^{(T^* + T)} = [1 + r(T^*)]^{T^*} \times [1 + f(T^*, T)]^T$
4. Spot rate for a security, having maturity of $T > 1$
 $r(T) = \{ [1 + r(1)] [1 + f(1,1)] [1 + f(2,1)] [1 + f(3,1)] \dots [1 + f(T-1,1)] \}^{(1/T)} - 1$

Forward rate model can be expressed as:

$$\left\{ \frac{[1 + r(T^* + T)]}{[1 + r(T^*)]} \right\}^{\frac{T^*}{T}} [1 + r(T^* + T)] = [1 + f(T^*, T)]$$

5. Yield Curve Movement and the Forward Curve
 - $F(T^*, T) = \frac{P(T^* + T)}{P(T^*)}$
 - $P^*(T) = \frac{P(t + T)}{P(t)}$
 - $F^*(t, T^*, T) = \frac{P^*(T^* + T - t)}{P^*(T^* - t)}$
 - $F^*(t, T^*, T) = \frac{\frac{P(t + T^* + T - t)}{P(t + T^* - t)}}{\frac{P(t)}{P(t + T^* - t)}} = \frac{P^*(T^* + T)}{P(T^*)} = F(T^*, T)$

Active Bond Portfolio Management

6. 1-yr. HPR = $\frac{[1 + r(T+1)]^{T+1}}{[1 + f(1, T)]^T} = [1 + r(1)]$
 (when the spot curve one year from today is today's forward curve)
7. Return of the 2-year zero-coupon bond over 1-yr HP =
 $\frac{\text{Price of a 2-yr zero-coupon bond 1 yr from today}}{\text{Purchase price of bond}} - 1$
 - Price of a 2-yr zero-coupon bond 1 yr from today =
 $\frac{\text{Parvalue of bond}}{(1 + \text{Fwd rate for 1 yr bond 1 yr from today})}$
 - Price of a 3-yr zero-coupon bond 1 yr from today =

$$\frac{\frac{\text{Parvalue of bond}}{(1 + \text{Fwd rate for 2 yr bond 1 year from today})}}{\frac{\text{Parvalue of bond}}{1 + f(1,2)}} =$$

8. Swap Spread = Fixed-rate of an interest rate swap – Interest rate on “on-the-run” Govt. security

$$\sum_{t=1}^T \frac{s(T)}{[1 + r(t)]^t} + \frac{1}{[1 + r(T)]^T} = 1$$

$\downarrow$ $\downarrow$
fixed rate leg *floating rate leg*
9. TED spread = LIBOR - T-bill rate of matching maturity
10. Libor-OIS spread = Libor - Overnight indexed swap (OIS) rate
11. Local expectations theory = $\frac{1}{P(t, T)} = \frac{1}{[1 + r(1)][1 + f(1,1)][1 + f(2,1)][1 + f(3,1)] \dots [1 + f(T-1,1)]}$
12. Cox-Ingersoll-Ross (CIR) Model = $dr = a(b - r) dt + \sigma \sqrt{rdz}$
13. Vasicek Model = $dr = a(b - r)dt + \sigma dz$
14. Ho-Lee model = $dr_t = \theta_t dt + \sigma dz_t$
15. Interest rate volatility for a security with maturity T at time t = $\sigma(t, T) = \frac{\sigma \left[\frac{\Delta r(t, T)}{r(t, T)} \right]}{\sqrt{\Delta t}}$

Reading 44: The arbitrage free valuation framework

Reading 45: Valuation & analysis: Bonds with embedded options

1. Value of callable bond = Value of straight bond – Value of issuer call option
2. Value of issuer call option = Value of straight bond – Value of callable bond
3. Value of putable bond = Value of straight bond + Value of investor put option
4. Value of investor put option = Value of putable bond – Value of straight bond
5. The rate in the up state = $R_u = R_d \times e^{2\sigma\sqrt{t}}$
where, R_d = Rate in the down state
 σ = Interest rate volatility
 t = Time in years between “time slices”
6. Duration = $\frac{V_- - V_+}{2 \times V_0 \times (\Delta Y)}$
7. Convexity = $\frac{V_+ + V_- - (2 \times V_0)}{2 \times V_0 \times (\Delta Y)^2}$
8. Effective Duration = $\frac{(PV_-) - (PV_+)}{2 \times (\Delta \text{Curve}) \times (PV_0)}$
9. Effective Convexity = $\frac{(PV_-) + (PV_+) - [2 \times (PV_0)]}{(\Delta \text{Curve})^2 \times (PV_0)}$
10. Value of capped floater = Value of straight bond – Value of embedded cap

11. Value of floored floater = Value of straight bond + Value of embedded floor

Analysis of a Convertible Bond

12. Conversion Ratio (CR) = No. of shares of C.stock from exercising call option
13. Conversion Price (or stated conversion price) = Par value of convertible bond ÷ CR
14. Conversion Value (or Parity) = Market price of C.stock × CR
15. Straight Value or Investment Value = Market value of a security without conversion option
16. Min. Value of a Convertible Security is (greater of conversion value or straight value)
17. Market Conversion Price or Conversion Parity Price = $\frac{\text{Market Price of Convertible Security}}{\text{CR}}$
18. Market Conversion Premium per share = Market Conversion Price – Current Market Price
19. Premium Payback Period = $\frac{\text{Market Conversion premium per share}}{\text{Favorable Income Differential per share}}$

20. Favorable Income Differential per share = $\frac{\text{Coupon interest} - (\text{CR} \times \text{C.stock Div. per share})}{\text{CR}}$

21. Premium over straight value = $\frac{\text{Market Price of Convertible Bond}}{\text{Straight Value}} - 1$

22. Non-callable/Non-putable Convertible security value = Straight value + Value of Call option on stock

23. Callable Convertible bond value = Straight value + Value of call option on stock – Value of call option on bond

24. Callable & Putable Convertible bond value = Straight value + Value of call option on stock – Value of call option on bond + Value of Put option on bond

25. Value of Call Option – Value of Put Option = PV (Forward price of bond on exercise date – Exercise price)

Reading 46: Credit analysis models

1. Expected loss = Full amount owed – Expected recovery or = Loss given default × Probability of default
2. Credit spread = Yield to maturity of a risky bond – Yield to maturity of a Govt. bond
3. Put option's price = Value of risky debt – Value of riskless debt

4. Black-Scholes Option pricing Formula =

$$S_t = A_t N(d_1) - ke^{-r(T-t)} N(d_2)$$

Where,

$$d_1 = \frac{\ln\left(\frac{A_t}{K}\right) + r(T-t) + \frac{1}{2}\sigma^2(T-t)}{\sigma\sqrt{T-t}}$$

$$d_2 = d_1 - \sigma\sqrt{T-t}$$

5. Value of debt = $D(t, T)$ = PV of payoff on co.'s debt if default occurs + PV of payoff on co.'s debt if default does not occur =

$$A_t N(-d_1) + Ke^{-r(T-t)} N(d_2)$$

- where, $N(d_2)$ = Risk neutral probability of the co.'s debt not defaulting

6. Credit Risk Measures

- Probability of the debt defaulting = Prob. ($A_T < K$) = $1 - \text{Prob.} (A_T \geq K) = 1 - N(e_2)$

Where,

$$e_1 = \frac{\ln\left(\frac{A_t}{K}\right) + u(T-t) + \frac{1}{2}\sigma^2(T-t)}{\sigma\sqrt{T-t}}$$

$$e_2 = e_1 - \sigma\sqrt{T-t}$$

7. Co.'s asset R in CAPM (a static one-period model):

Co.'s asset $R = R_f + \beta$ of co.'s asset
(Expected R per year on Mkt. portfolio – R_f)
= $R_f + (\beta \text{ of co.'s asset} \times \text{Mkt.'s ERP})$

8. Price of debt $D(t, T) =$
$$\bar{E} \left[\frac{K}{(1+r_1\Delta)(1+r_{1+\Delta})\dots(1+r_{T+\Delta})} \right]$$

9. Credit risk measures in reduced form model:

- Default probability over $[0, T]$ = Prob ($t \leq T$) = $1 - E \left\{ \frac{t(X_t)K}{[1+\lambda(X_0\Delta)]+[1+\lambda(X_\Delta)]\dots[1+\lambda(X_{T-\Delta})\Delta]} \right\}$
- Expected loss =
$$E_{t=0}^{T-\Delta} \left\{ \frac{t(X_t)K}{[1+\lambda(X_0\Delta)]+[1+\lambda(X_\Delta)]\dots[1+\lambda(X_t)\Delta]} \lambda(X_t)\Delta \right\}$$
- Present value of the expected loss = $K [P(t, T) - D(t, T)]$

10. Historical Estimation

- Probability of default over $[t, t+\Delta] =$
$$\text{Prob}(t) = \frac{1}{1 + e^{-\alpha - \sum_{i=1}^N b_i X_t^i}}$$
- Parameters estimation:
$$\ln\left(\frac{dt}{1-dt}\right) = \alpha + \sum_{i=1}^N b_i X_t^i$$
 Where
 $dt = \{1 \text{ if default, } 0 \text{ if no default}\}$

- To estimate the loss given default:

$$t(X_t) = c_0 + \sum_{i=1}^N c_i X_t^i$$

Where $\{c_i \text{ for } i = 1, \dots, N\}$ are constants.

11. Price of the coupon bond (assuming no arbitrage and frictionless markets)

$$B_G(t) = \sum_{i=1}^{T-1} CP(t, i) + (C + F)P(t, T)$$

12. Credit spread (t) = Avg. yields on risky zero-coupon bond – Avg. yields on riskless zero-coupon bond

Or

= [Average yields on the risky zero-coupon bond – Average yields on riskless zero-coupon bond] + Liquidity premium

or

= Expected % loss per year on the risky zero-coupon bond + Liquidity Premium

13. PV of expected loss = PV of CF of riskless debt – PV of CF of risky debt = $[P(t, T) - D(t, T)] X_T$

Where, X_T = Promised CF at T of a risky Co.

Reading 47: Forward Rate Markets & Contracts

1. Forward Price = Spot price $\times (1 + R_f)^T$
 $F(0, T) = S_0 (1 + r)^T$ Or

$$S_0 = F(0, T) / (1 + r)^T$$

2. Value of a forward contract (of long position) at time t during contract life (where $t < T$):

$$V_t(0, T) = S_t - F(0, T) / (1 + r)^{(T-t)}$$

3. Value of a forward contract (of Short position) at time t during contract life (where $t < T$):

$$V_t(0, T) = F(0, T) / (1 + r)^{(T-t)} - S_t$$

4. Value of a Forward contract (of long position) at expiration ($t = T$):

$$V_T(0, T) = S_T - F(0, T)$$

5. Value of a Forward contract at initiation (of long position) ($t = 0$):

$$V_t(0, T) = S_0 - F(0, T) / (1 + r)^T = 0$$

6. No-arbitrage price of an equity forward contract = $F(0, T) = (S_0 - PVD) \times (1 + r)^T$

$$\text{Or } F(0, T) = [S_0 \times (1 + r)^T] - FVD$$

7. Value of the forward contract on an equity (of long position):

$$V_T(0, T) = [S_t - PVD_t] - [F(0, T) / (1 + r)^{(T-t)}]$$

8. Equity Forward Contracts With Continuous Dividends = $F(0, T) = (S_0$

$$e^{-\delta^c T}) e^{r^c T}$$

$$\text{Or } F(0, T) = S_0 \times e^{(r^c - \delta^c)T}$$

9. For the continuous time case, forward contract value (of long position)

$$V_T(0, T) = S_t e^{-\delta^c(T-t)} - F(0, T) e^{-r^c(T-t)}$$

Or

V_t (of the long position)

$$= \left[\frac{S_t}{e^{\delta^c \times (T-t)}} \right] - \left[\frac{FP}{e^{r^c \times (T-t)}} \right]$$

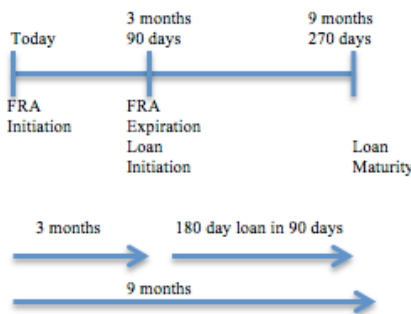
10. Price of Forward Contracts on Fixed Income = $F(0, T) = (S_0 - PVC) \times (1 + r)^T$

$$\text{Or } F(0, T) = [S_0 \times (1 + r)^T] - FVC$$

11. The value of the forward contract (of long position) prior to expiration:

$$V_T(0, T) = [S_t - PVC_t] - [F(0, T) / (1 + r)^{(T-t)}]$$

12. Forward Rate Agreements (FRAs): An example of 3 × 9 FRA (read as three by nine):



- Contract expires in 90 days

- Underlying loan settled in 270 days
- Underlying rate is 180-day LIBOR

13. Covered interest rate parity = $F(0, T) = S_0 \times [(1 + r)^T \div (1 + r^f)^T]$

14. Valuing Currency Forward Contracts = $V_t(0, T) = [S_0 \div (1 + r^f)^{(T-t)}] - [F(0, T) \div (1 + r)^{(T-t)}]$

15. Continuous time price formulas for currency forward contracts:

$$F(0, T) = [S_0 e^{-r^f T}] e^{r^c T}$$

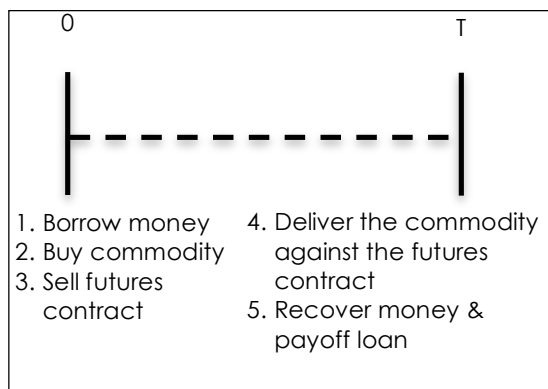
$$\text{Or } F(0, T) = S_0 \times e^{(r^c - r^f)T}$$

16. Continuous time value formulas for currency forward contracts:

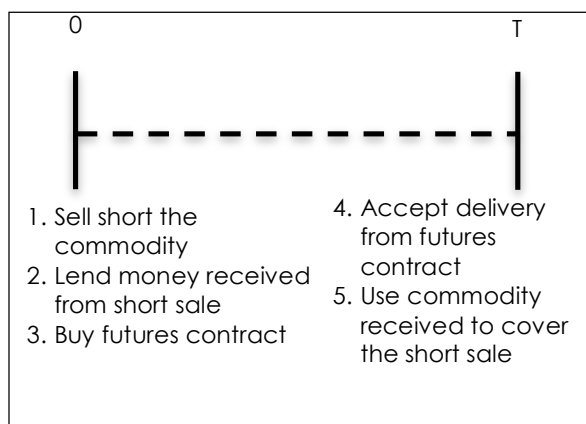
$$V_t(0, T) = (S_t e^{-r^f(T-t)}) - F(0, T) e^{-r^c(T-t)}$$

Reading 48: Futures Markets & Contracts

- Basis = Spot Price – Futures Price
- Futures price for an asset with no storage costs = $\text{Spot Price} \times (1 + R_f)^{\text{Time}}$
 $F(0, T) = S_0 (1 + r)^T$ Or $S_0 = F(0, T) / (1 + r)^T$
- C&C Arbitrage (when the Futures Contract is Overpriced)



4. R C&C Arbitrage (when the Futures Contract is Underpriced)



5. Pricing Futures Contracts when there are Storage Costs and/or cash flows on the Underlying asset and/or convenience yield
- If holding an underlying asset results in monetary costs and benefits (net cost), futures price is:

$$FP = S_0 \times (1 + R_f)^T + FV(NC)$$

- If holding an underlying asset results in non-monetary benefits (convenience yield), futures price is:

$$FP = S_0 \times (1 + R_f)^T - FV(NB)$$

6. Pricing T-Bond futures Contract = $F(0, T) = (S_0 - PVC) \times (1 + r)^T$
Or
 $F(0, T) = [S_0 \times (1 + r)^T] - FVC$

7. No-arbitrage price of a Stock index futures contract = $F(0, T) = (S_0 - PVD) \times (1 + r)^T$
Or $F(0, T) = [S_0 \times (1 + r)^T] - FVD$

8. Futures price as stock index discounted at dividend yield, compounded at R_f rate:
 $F(0, T) = [S_0 / (1 + \delta)^T] \times (1 + r)^T$
Or $F(0, T) = S_0 (1 - \delta^*) (1 + r)^T$

9. Stock Index Futures Contracts with Continuous Dividends = $F(0, T) = (S_0 e^{-\delta^* T}) e^{r^* T}$ Or
 $F(0, T) = S_0 e^{(r^* - \delta^*) T}$

10. No-arbitrage futures price of a unit of foreign currency in terms of the home currency for a currency futures contract of length T in years = $F(0, T) = S_0 \times [(1 + r)^T / (1 + r^f)^T]$

11. The continuous time price formulas for currency futures contracts:

$$F(0, T) = \left[S_0 e^{-r^f T} \right] e^{r^c T} \text{ or } S_0 \times e^{(r^c - r^f) T}$$

Reading 49: Options Markets & Contracts

1. Put-call parity = $c_0 + X / (1 + R_f)^T = p_0 + S_0$
- European call (synthetic call) = $c_0 = p_0 + S_0 - X / (1 + R_f)^T$
 - European put (synthetic put) $p_0 = c_0 - S_0 + X / (1 + R_f)^T$
 - Synthetic stock position = $S_0 = c_0 - p_0 + X / (1 + R_f)^T$
2. Delta = hedge ratio = number of shares purchased per call sold
- No. of stocks per option (D) = $(C^+ - C^-) / (S^+ - S^-)$
 - No. of options per stock = $1/D$
 - No. of calls to sell = No. of shares hedged / delta of the call option
 - Total shares of stock to purchase = No. of options short $\times$ Delta

The two values of the put at the end of the first period are

$$P_u = \frac{pP_{uu} + (1-p)P_{ud}}{1+r}$$

$$P_d = \frac{pP_{du} + (1-p)P_{dd}}{1+r}$$

$$3. \text{ Expiration value of caplet} = \frac{\max[0, \{(1\text{-yr rate} - \text{cap rate}) \times \text{NP}\}]}{1 + \text{one-yr rate}}$$

$$4. \text{ Expiration value of floorlet} = \frac{\max[0, \{(\text{floor rate} - 1\text{-yr rate}) \times \text{NP}\}]}{1 + \text{one-yr rate}}$$

5. The Black-Scholes equation:

$$c = S_0 N(d_1) - X e^{-r^c T} N(d_2)$$

$$p = X e^{-r^c T} [1 - N(d_2)] - S_0 [1 - N(d_1)]$$

$$\text{where } d_1 = \frac{\ln(S_0 / X) + [r^c + (\sigma^2 / 2)]T}{\sigma \sqrt{T}}$$

$$d_2 = d_1 - \sigma \sqrt{T}$$

6. Delta = $\partial c / \partial S = N(d_1)$

where, ∂S = a small Δ in stock price;

∂c = the corresponding Δ in call price.

- Call option delta = $N(d_1)$ and Put option delta is $N(d_1) - 1$.
- Δ in option price = Delta $\times \Delta$ in underlying price
- $\Delta C \approx N(d_1) \times \Delta S$
- $\Delta P \approx [N(d_1) - 1] \times \Delta S$

7. Put-call parity for options on underlying assets with CF by adjusting S for PVCF:

$$c_0 + X / (1 + R_f)^T = p_0 + (S_0 - \text{PVCF})$$

8. Historical volatility estimate = S.D of $R =$

$$\sqrt{\frac{1}{n-1} \sum_{i=1}^N (R_i - \text{Avg. } R_i)^2}$$

9. Annualized Historical Volatility (when daily returns are used) = $H.V \times \sqrt{252}$

10. The Black Model equation for European option on futures and forward contracts:

$$c = e^{-r^c T} [f_0(T) N(d_1) - X N(d_2)]$$

$$p = e^{-r^c T} (X[1 - N(d_2)] - f_0(T)[1 - N(d_1)])$$

$$d_1 = \frac{\ln(f_0(T) / X) + 0.5\sigma^2 T}{\sigma \sqrt{T}}$$

$$d_2 = d_1 - \sigma \sqrt{T}$$

Reading 50: Swap Markets and Contracts

1. Pricing and Valuation of Swaps

- The fixed periodic-rate on an N -period swap at initiation:

$$R_N = \frac{1 - Z_N}{Z_1 + Z_2 + \dots + Z_N}$$

$$Z_n = \frac{1}{1 + (L_n \times \text{days} / 360)}$$

- $V_{\text{fixed rate}} = [\text{Fixed payment} \times (Z_1 + Z_2 + \dots + Z_N)] + (\$1 \times Z_N)$
- $V_{\text{floating rate}} = (\$1 + 1^{\text{st}} \text{ floating pmt}) \times Z_1$

$$V_{\text{Equity}} = \$1 \times \left[\frac{\text{Index}(A)_t}{\text{Index}(A)_0} - \frac{\text{Index}(B)_t}{\text{Index}(B)_0} \right]$$

2. Pricing and Valuation of Swaps:

- Payer swaption (payoff) = $\text{Max} [0, (\text{Mkt rate} - X \text{ rate})] \times \sum (\text{discount factors})$
- Put option (payoff) = $\text{Max} [0, (1.00 - \text{Mkt value of coupon bond})]$
- Receiver swaption (payoff) = $\text{Max} [0, (X \text{ rate} - \text{Mkt rate})] \times \sum (\text{discount factors})$
- Call option (payoff) = $\text{Max} [0, (\text{Mkt value of coupon bond} - 1.00)]$

3. Swap Spread = Fixed-rate on the Swap – Yield on default-free security of the same maturity as the swap

Reading 51: Interest Rate Derivative Instruments

1. Payoff to Cap Buyer = $\text{Max} [0, \text{NP} \times (\text{Ref. rate} - \text{cap rate}) \times (\text{actual days} / 360)]$.
2. Payoff to Floor Buyer = $\text{Max} [0, \text{NP} \times (\text{floor rate} - \text{ref. rate}) \times (\text{actual days} / 360)]$

Reading 52: Credit Default Swaps

1. Upfront premium = Credit spread – Standard rate

2. Expected Credit Loss (%) = Payout ratio = $1 - \text{Recovery rate (\%)}$
3. Expected Credit Loss Amount or Payout amount = Payout ratio $\times$ Notional amount
4. Loss Given Default:
 - Expected loss = Full amount owed – Expected recovery
 - Expected loss = Loss given default $\times$ Probability of default
 - Prob. of default (at some point during T years) = $1 - \text{Prob. of no default during T years}$
5. Value of protection leg = Expected payoff of bond/loan with credit risk - Expected payoff of bond/loan with no credit risks
6. Value of premium leg = PV of pmts. made by the protection buyer to the protection seller
7. Upfront pmt = PV of protection leg – PV of premium leg
8. Credit spread $\approx$ Prob. of default $\times$ Loss given default (%)
9. Credit spread Pricing Conventions
 - Upfront premium = PV of credit spread – PV of fixed coupon Or = (Credit spread – Fixed coupon) $\times$ D of the CDS
- PV of credit spread = Upfront prem. + PV of fixed coupon
- Credit spread $\approx$ (Upfront prem./D) + Fixed coupon
- Upfront premium in % = $100 - \text{Price of CDS in currency per 100 par}$
- Price of CDS in currency per 100 par = $100 - \text{Upfront premium \%}$
10. Profit for the buyer of protection $\approx \Delta$ in spread in bps $\times$ D $\times$ NP
11. % change in CDS price = Δ in spread in bps $\times$ D
12. Basis = CDS spread (prem.) – Bond's credit spread*
 *Bond's Credit spread = Yield on bond - Investor's cost of funding
 Bond yield = R_f rate + Funding spread + Credit spread
 where, R_f + Funding spread = LIBOR
13. Synthetic CDO = Portfolio of default-free securities + CDS holdings
3. Carhart Four Factor Model = $E(R_p) = R_f + \beta_{p1}RMRF + \beta_{p2}SMB + \beta_{p3}HML + \beta_{p4}WML + \epsilon_p$

 RMRF = Portfolio's sensitivity to Mkt. Index
 SMB = small minus big
 HML = high minus low
 WML = winners minus losers
4. Macroeconomic Factor Model = $R_i = a_i + b_{i1}F_1 + b_{i2}F_2 + \dots + b_{iK}F_K + \epsilon_i$
5. $b_{iK} = \frac{\text{Value of } k \text{ for asset } i - \text{Average value of } k}{\sigma(\text{for values of } k)}$
6. Actual Inf. = Predicted Inf. + Surprise Inf.
7. Active R = $R_p - R_B$
8. Active R (decomposition) = $\sum_k^1 [(\text{Portfolio sensitivity}) - (\text{benchmark sensitivity})_k] \times (\text{Factor } R)_k + \text{security selection}$
9. Tracking Error TE = $s(R_p - R_B)$
10. Information Ratio = $\frac{\bar{R}_p - \bar{R}_B}{s(R_p - R_B)}$
11. Active risk squared = $s^2(R_p - R_B)$
12. Active risk squared = Active factor risk + Active specific risk

Reading 53: An Introduction to Multifactor Models

1. Multifactor Model = $R_i = a_i + b_{i1}I_1 + b_{i2}I_2 + \dots + b_{iK}I_K + \epsilon_i$
2. Arbitrage Pricing Theory = $E(R_p) = R_f + \lambda_1\beta_{p,1} + \lambda_2\beta_{p,2} + \dots + \lambda_k\beta_{p,k}$

$$13. \text{ Active specific risk} = \sum_{i=1}^n (w_i^a)^2 \sigma_{\epsilon_i}^2$$

Where

w_i^a = ith asset's active weight

$\sigma_{\epsilon_i}^2$ is ith asset's residual risk

Reading 54: Analysis of Active Portfolio Management

$$1. \text{ Benchmark Portfolio} = R_B = \sum_{i=1}^N w_{B,i} R_i$$

$$2. \text{ Portfolio Return} = R_P = \sum_{i=1}^N w_{P,i} R_i$$

$$3. \text{ Value added return} = R_P - R_B$$

$$4. \text{ Sharpe Ratio } SR = \frac{R_P - R_F}{S.D.(R_P)}$$

$$5. \text{ Information Ratio } IR = \frac{R_P - R_B}{S.D.(R_P)} = \frac{R_A}{S.D.(R_P)}$$

$$6. SR_P^2 = SR_B^2 + IR^2$$

$$7. S.D.(R_A) = \frac{IR}{SR_B} \times S.D.(R_B)$$

$$8. \text{ Active Security R as residual R in}$$

- Single factor statistical model $= R_{Ai} = R_i - R_B$
- Multi factor statistical model $= R_{Ai} = R_i - \sum_{j=1}^k \beta_{j,i} R_j$

$$9. \text{ Mean-var optimal security wghts for uncorrelated active R subject to limit on active portfolio risk} =$$

$$\Delta w_i^* = \frac{\mu_i}{\sigma_i^2} \frac{\sigma_A}{\sqrt{\sum_{i=1}^N \frac{\mu_i^2}{\sigma_i^2}}}$$

$$10. \text{ Grinold Rule} = \mu_i = IC \sigma_i S_i$$

$$11. \Delta w_i^* = \frac{\mu_i}{\sigma_i^2} \frac{\sigma_A}{IC \sqrt{BR}}$$

$$12. \text{ Anticipated value added for Active portfolio} = E(R_A) = IC \sqrt{BR} \sigma_A$$

$$13. \text{ Transfer Coefficient } TC = \text{Cor}(\mu_i / \sigma_i, \Delta w_i \sigma_i)$$

$$14. \text{ Fundamental Law: } E(R_A) = (TC)(IC) \sqrt{BR} \sigma_A$$

$$15. \sigma_A = TC \frac{IR^*}{SR_B} \sigma_B$$

$$16. SR_P^2 = SR_B^2 + (TC)^2 (IR^*)^2$$

Ex Ante Measurement of Skill

$$17. \sigma_A = \sigma_{IC} \sqrt{N} \sigma_{RM}$$

$$18. E(R_A) = \frac{IC}{\sigma_{IC}} \sigma_A$$

$$19. \text{ Breadth } BR = \frac{N}{1 + (N-1)\beta}$$

Reading 55: Economics & Investment Markets

$$1. \text{ Present Value Model} =$$

$$\sum_{s=1}^N \frac{E_t[\widetilde{CF}_{t+s}]}{(1 + l_{t,s} + \theta_{t,s} + \rho_{t,s}^i)^s}$$

$$2. \text{ Price of default-free bond certain to pay off one unit of real consumption at time } s = P_{t,s} = E_t[1 \widetilde{m}_{t,s}] = E_t[\widetilde{m}_{t,s}]$$

$$3. \text{ One period } R_f \text{ interest rate } l_{t,1} = \frac{1 - P_{t,1}}{P_{t,1}} = \frac{1}{E_t[\widetilde{m}_{t,1}]} - 1$$

Risk premium on risky assets:

$$4. \text{ Relation b/w expected value and Cov.} = E_t(\widetilde{x} \widetilde{y}) = E_t(\widetilde{x}) E_t(\widetilde{y}) + \text{Cov}(\widetilde{x}, \widetilde{y})$$

$$5. \text{ Alternate way to view the pricing relation} = P_{t,s} = \frac{E_t[\widetilde{P}_{t+1,s-1}]}{1 + l_{t,1}} + \text{cov}_t[\widetilde{P}_{t+1,s-1}, \widetilde{m}_{t,1}]$$

$$6. \text{ Expected Holding period return} = r_{t,s} = \frac{E_t[\widetilde{P}_{t+1,s-1}] - P_{t,s}}{P_{t,s}}$$

$$7. \text{ Pricing for real default free } i \text{ rate} = P_t^i = \sum_{s=1}^N \frac{CF_{t+s}^i}{(1 + l_{t,s})^s}$$

Reading 56: The Portfolio Management Process & the Investment Policy Statement